A Software Engineering Project
On

Blood Bank & Donation Management System

as the partial fulfillment of

Semester -V study

for the degree of

Bachelor of Science (Computer Science)
(Affiliated to Veer Narmad South Gujarat University)

during the academic year 2022-2023.

Developed By:

Devani Yagnik

Guided By:

Dr. Priti Patel

Sarvajanik Education Society
Shree Ramkrishna Institute of Computer Education and Applied Sciences
Behind P.T Science College, M.T.B. College Campus, Athwalines, Surat-395 001.

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Shree Ramkrishna Institute of Computer Education and Applied Sciences
Behind P.T. Science College, M.T.B College Campus, Athwalines, Surat-395 001.

CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE

This is to certify that *Exam no(spid)*

1) Devani Yagnik 20192 (E20110192000310026)

has / have successfully completed his / her / their software engineering project work entitled

Blood Bank & Donation Mangement System

as the partial fulfillment of

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during the academic year 2022-2023.

_____ _____

Internal Project Guide **HOD**
Dept. of Computer Science

Date : _____ **Place : Surat**

University Examination:

Examination No. : _____

Signature of External Examiners : _____

_____

_____

INDEX for Project Documentation

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1. INTRODUCTION

1.1 System Profile

- Project title : Blood & Bank Donation Management System
- Front-end : HTML , CSS , JS
- Back-end : PHP , MySQL
- Submitted By : Devani Yagnik N
- Project guided : By Dr. Priti Patel
- Organization : Shree Ramkrishna Institute of Computer Education & Applied Science

2. SYSTEM INTRODUCTION

2.1 System Definition, Objective and scope

2.1.1 System Information: -

- is designed in such a manner that it can suit the needs of all the blood bank requirements in The project blood bank management system is known to be a pilot project that is designed for the blood bank to gather blood from various sources and distribute it to needy people who have high requirements for it.

- The Software is designed to handle the daily transaction of the blood bank and Search the details when required.

- It also helps to register the details of donors, blood collection details, as well as blood, issued reports

- The software Application the course of the future.

- It will help us to find the Blood group with its most efficient time to take care of the blood and it is easier to hand over the blood to the hospital to help people to get blood on time.

- This all things have been stored and seen in this blood bank management system. To help more people try their best to do so.

2.1.2 Project Objective: -

- The main objective of the Blood Bank Management System is to manage the details of Blood, Donor, and Blood Group.
- It manages all the information about Blood.
- project is built at the administrative end and thus only the administrator is guaranteed the access
- Admin and user can operate the system 24/7.
- To check the availability of blood bags anytime
- To manage the information of its blood donor
- Support fast searching to find match blood bags for the right person.

2.1.3 System Scope: -

- Login function accessed by only admin .
- Admin can only delete user.

2.2 Hardware & Software Requirements

Hardware:-

- Processor : Intel core i3 or above
- Ram : 2 GB or above
- Hard-Disk : 100 GB or above

Software:-

- Backend : PHP & My Sql (PHP MY Admin).
- Frontend : HTML , CSS , JS
- Operating System : Windows 8 or above

3. REQUIREMENT ANALYSIS

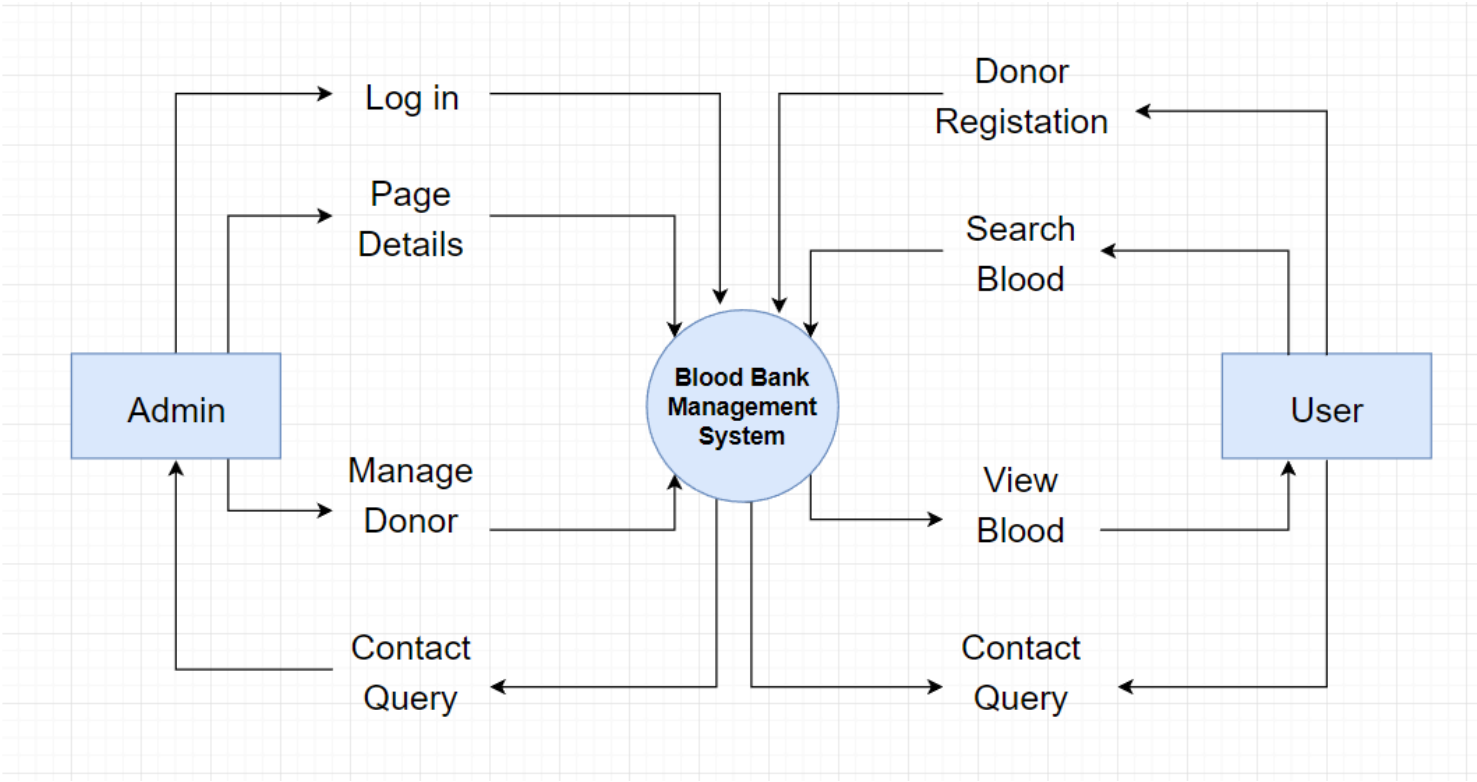
3.1 Expected working of System

- **Admin Module :** Admin can login with username and password which is already store in System database.
  - Login Facility
  - Donor Manage
  - Check Contact Query
  - Camp Manage
  - Pages Manages
  - Change Password
  - Logout

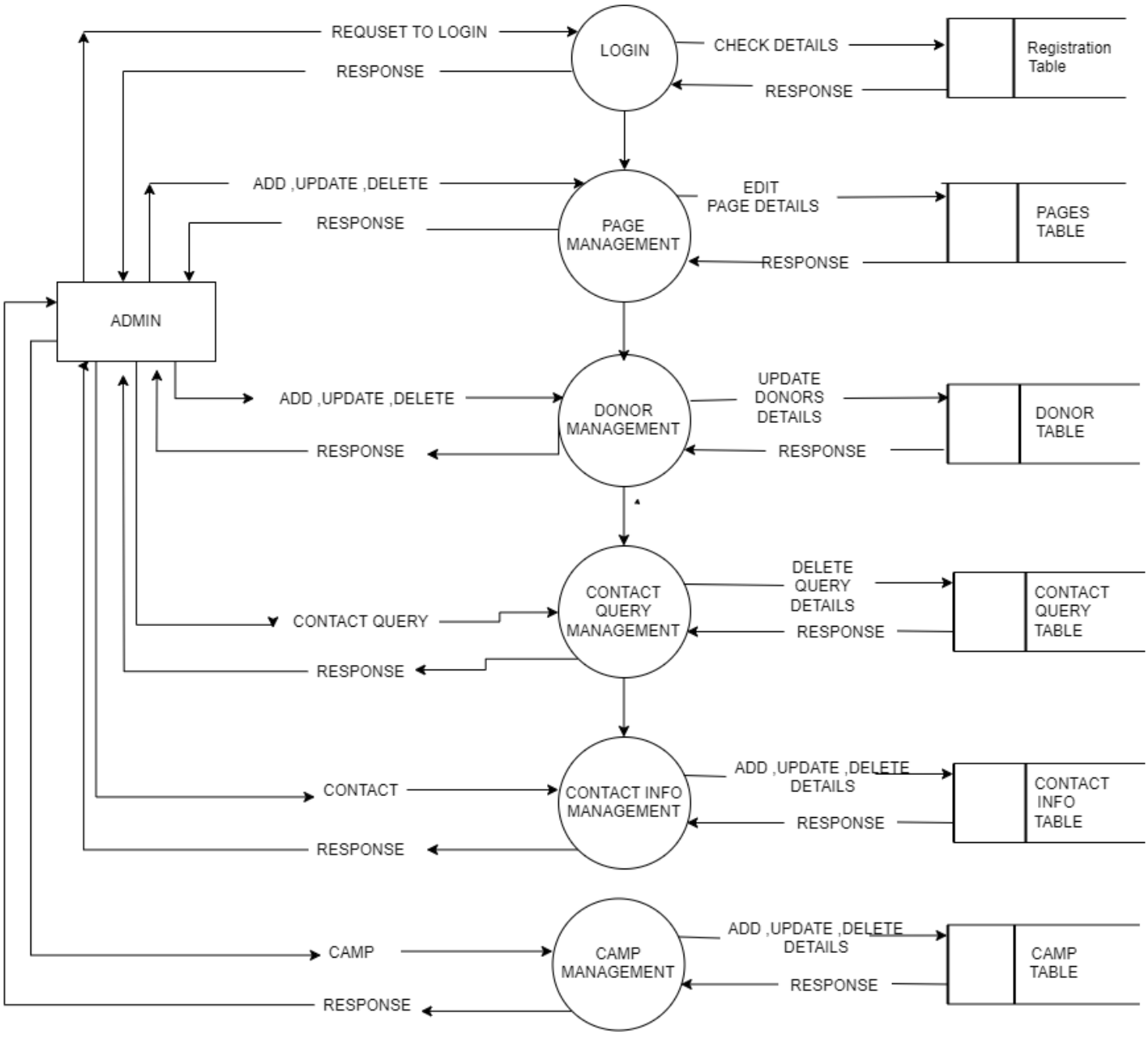
- **User Module :** User can donate blood and search blood donor as per blood group.
  - View Pages
  - View And Enroll Camp Details
  - View Blood Donor Info
  - Registration as Donor
  - Search Available Blood
  - Contact Admin

3.2 Data Flow Diagrams (DFD)

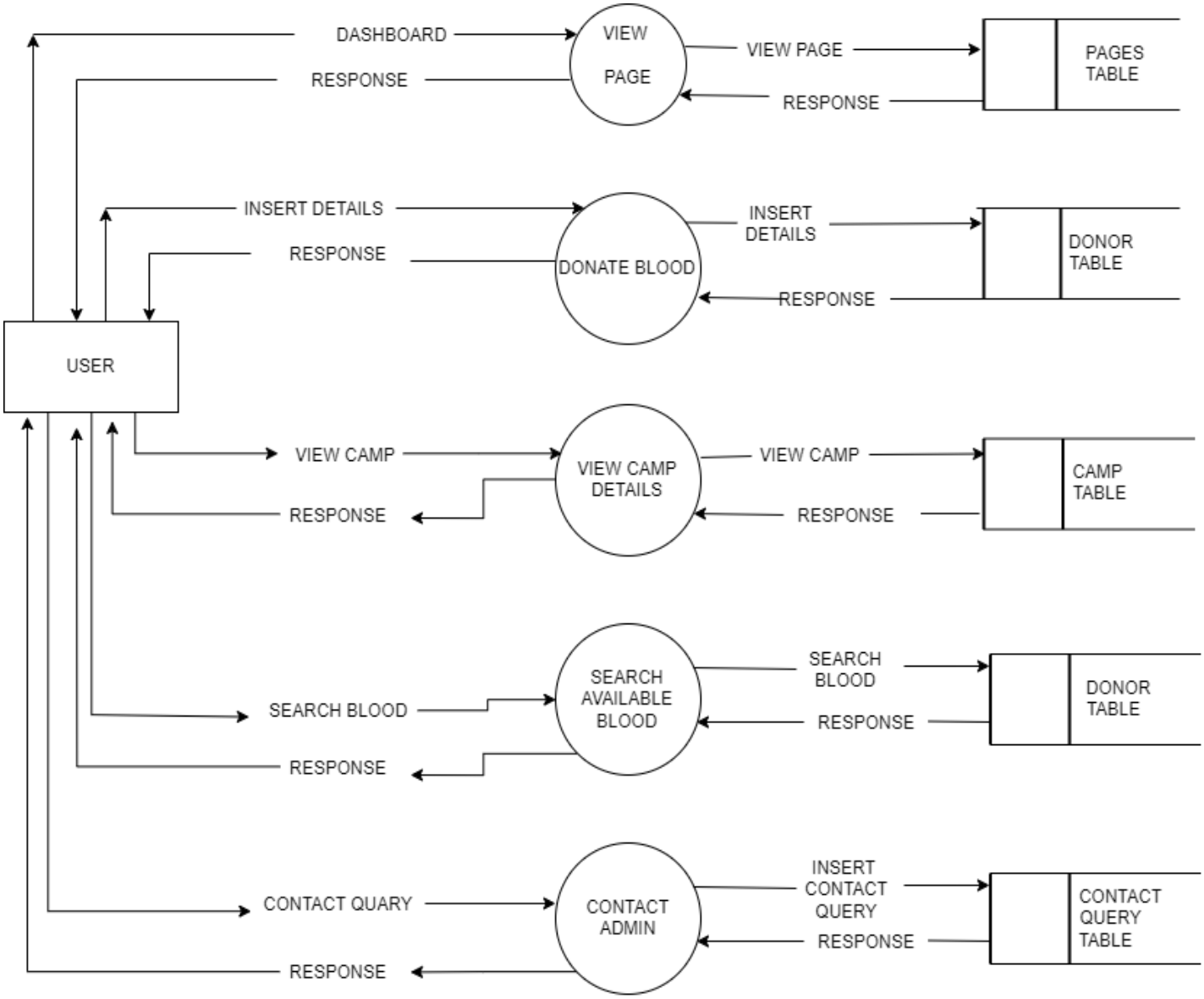
0 LEVEL DFD:-



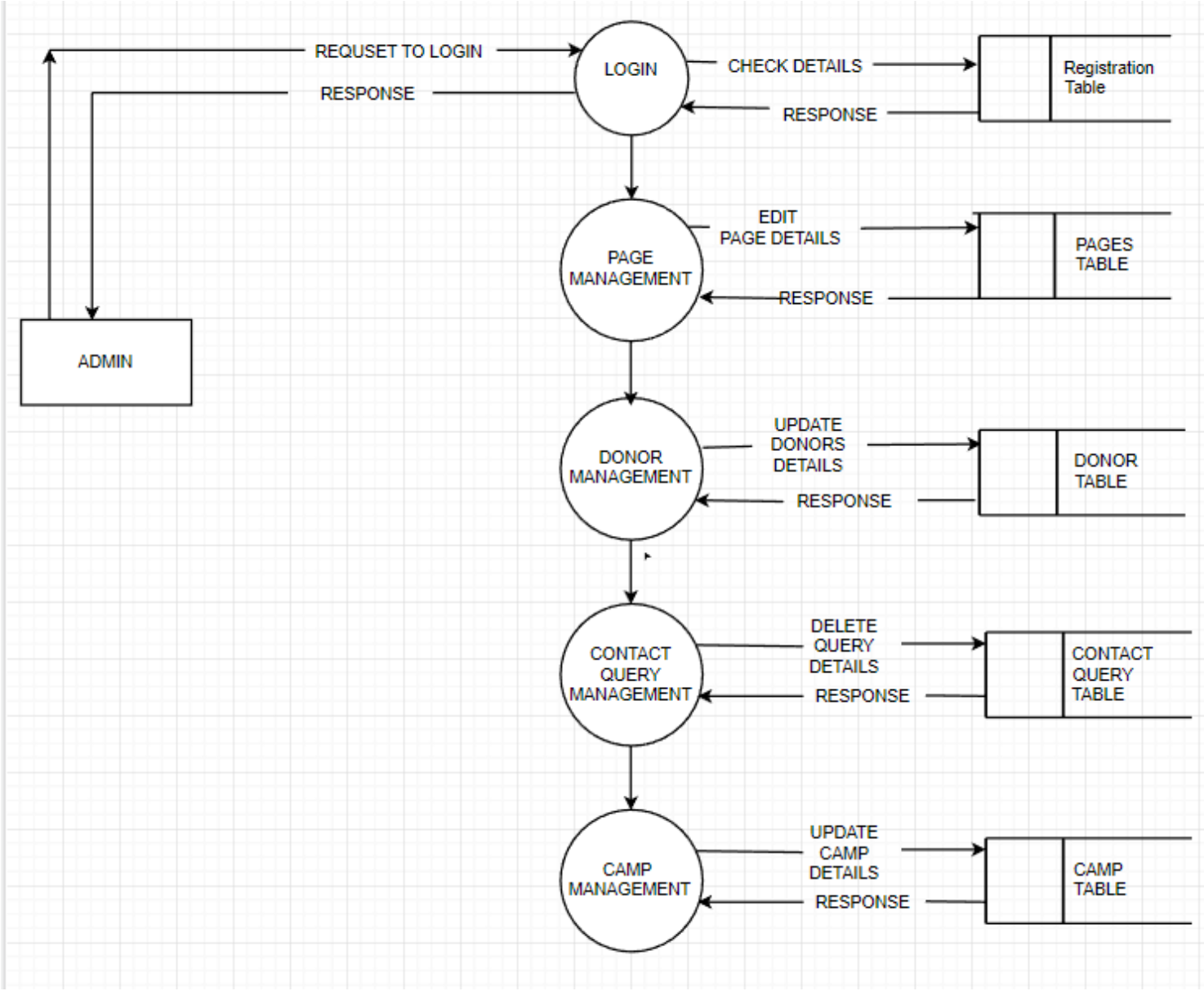
- **1st LEVEL DFD FOR ADMIN :-**



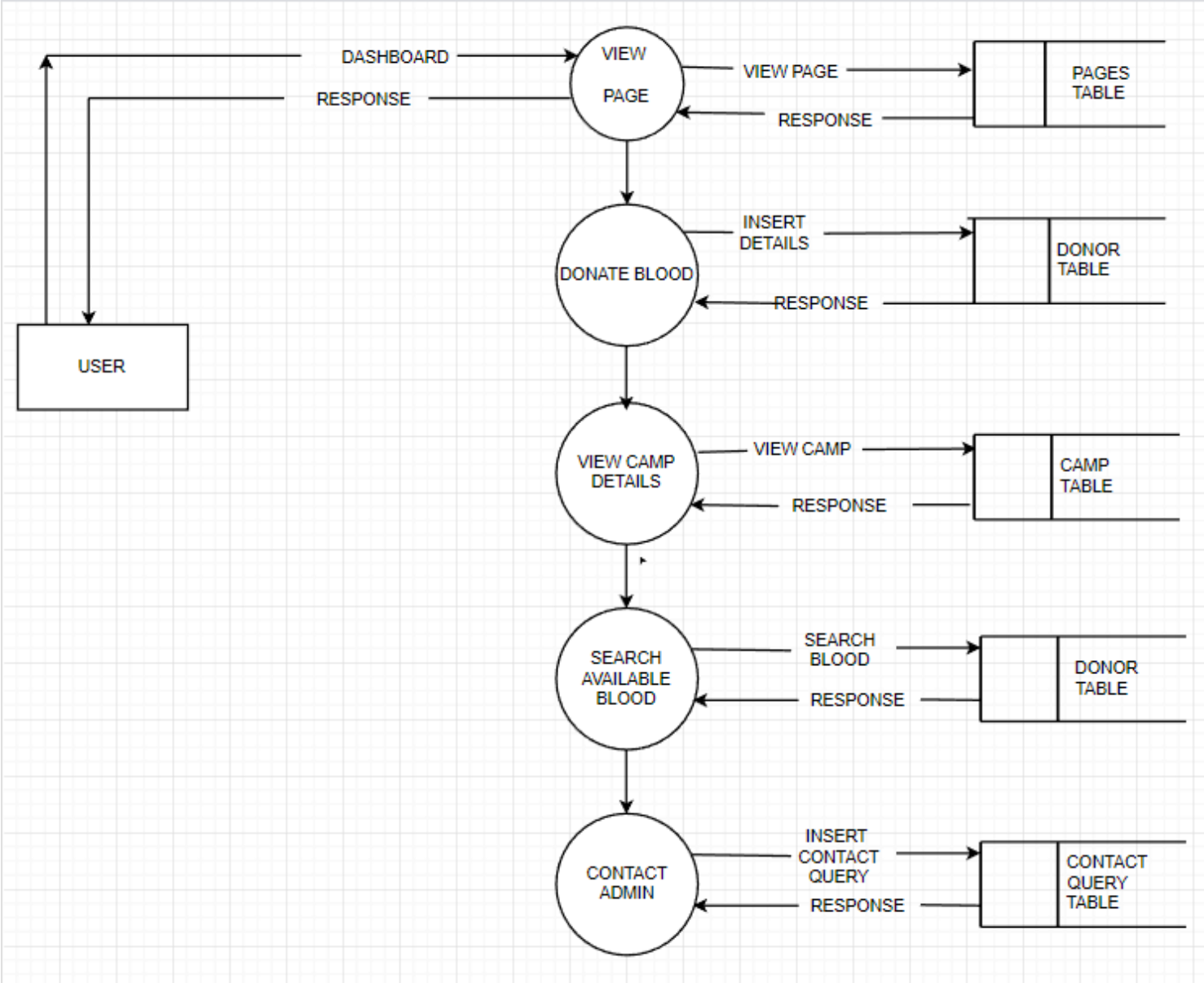
- **1st LEVEL DFD FOR ADMIN**



- **2nd LEVEL DFD FOR ADMIN :-**



- **2ST LEVEL DFD FOR USER :-**



3.3 Process Specification

▶ PROCESS 1.0 :

Name:	**Login**
Description:	This process is save all the information of the trust manager and Admin like Username , Password etc.

▶ PROCESS 1.0 FOR DONOR:

Name:	**Registration**
Description:	This process is save all the information of the trust manager or camp manager and donor like Username , Password etc.

▶ PROCESS 2.0 FOR ADMIN:

Name	**Manage User datails**
Description:	Admin can update and delete donor details

▶ PROCESS 3.0 FOR ADMIN:

Name:	**Manage Subject**
Description:	Admin can add camp and also delete camp

▶ PROCESS 4.0 FOR Donor:

Name:	**Fill query form**
Description:	Donor can fill the query form.

▶ PROCESS 5.0 FOR User:

Name:	**Search Available Blood**
Description:	Users can search for available blood

▶ PROCESS 6.0 Blood Donation:

NAME:	Blood Donation
Description:-	New donor ADD DATA Internet

3.4 Data Dictionary

- **Admin Table**

Name:	Admin Details
Alias:	Admin login
Where used/How used:	Data used for admin login
Content Description:	Admin id Admin Name Username password

- **Blood Table**

Name:	Blood details
Alias:	System
Where used/How used:	Data use for a particular Blood name with Id
Content Description:	Blood id Blood Group

- **Camp Table**

Name:	Camp details
Alias:	Camp
Where used/How used:	Camp Details for Admin and User
Content Description:	Camp id Date Camp Name Address Contact Conducted By Organized By Time

- **Enroll Camp Table**

Name:	Camp Enroll Donor Details
Alias:	Camp
Where used/How used:	Data use for details of Enroll Donor
Content Description:	Id Camp Name Donor name Number Email Age Gender Blood Group Address Camp id

- **Contact Information Table**

Name:	Contact Information
Alias:	Contact
Where used/How used:	Data use for post Contact view on user side
Content Description:	Id Address Mail Phone

- **Contact Query Table**

Name:	Contact Query
Alias:	Solve problem
Where used/How used:	Data use for details of donor query
Content Description:	id name mail number message date

- **Donor Table**

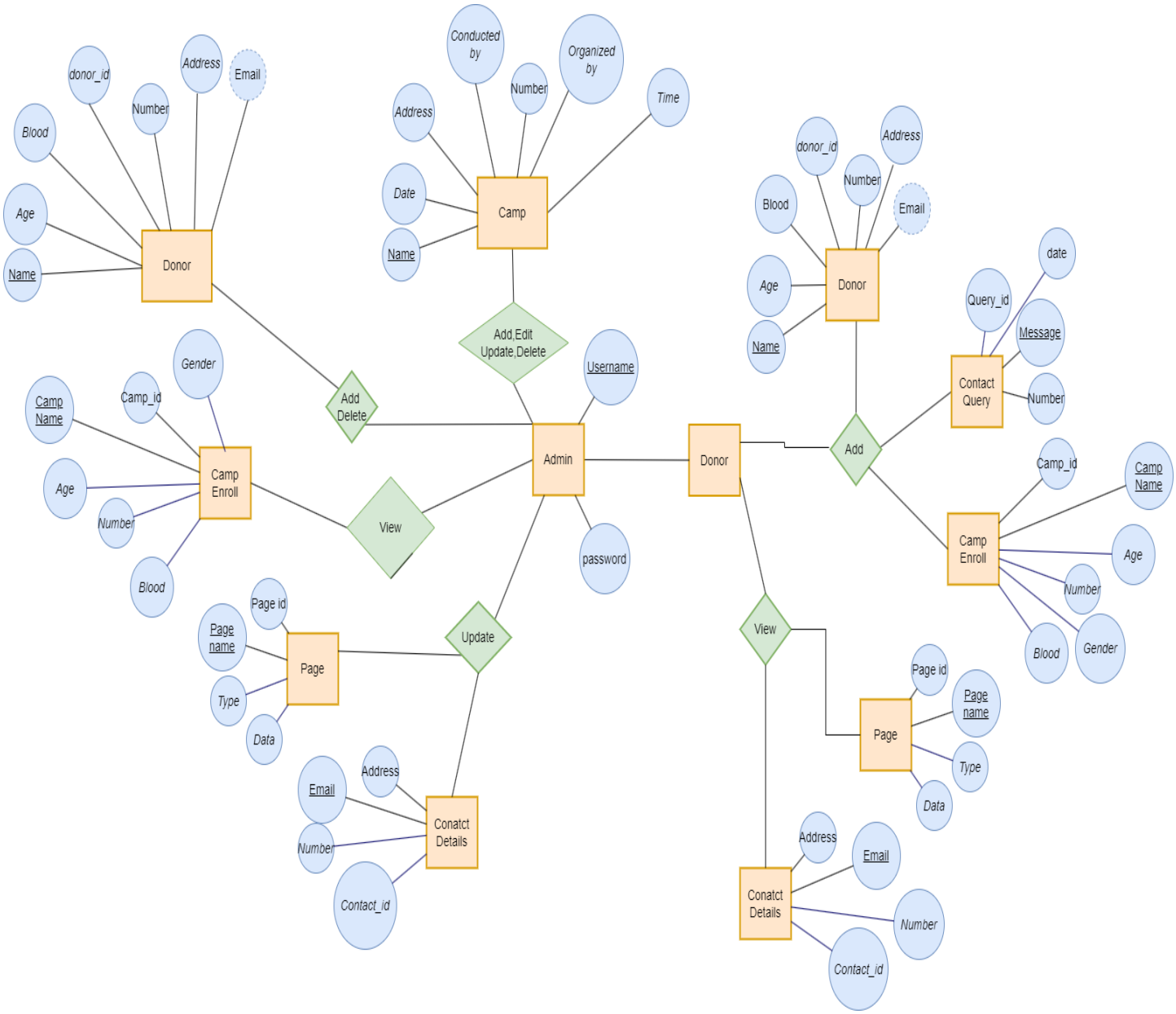
Name:	Donor details
Alias:	Donor
Where used/How used:	Donor Details use for Blood Donation
Content Description:	Id Name Number Mail Age Gender Blood Address

- **Page Table**

Name:	Page details
Alias:	page
Where used/How used:	Page Details for User
Content Description:	Id Page Name Page Type Page Data

4. Design

4.1 ER Diagram :



4.1 Database Design

- **admin_info Table**

Field Name	Data Type	Size	Constraint
Admin_id	int	10	Primary Key, Auto increment
Admin_name	text	50	Not Null
Admin_username	varchar	50	Not Null
Admin_password	varchar	50	Not Null

- **blood Table**

Field Name	Data Type	Size	Constraint
blood_id	int	11	Primary Key, Auto increment
blood_group	text	10	Not Null

- **camp Table**

Field Name	Data Type	Size	Constraint
camp_id	int	10	Primary Key, Auto increment
camp_name	text	50	Not Null
Date	Date	50	Not Null
Address	Varchar	50	Not Null
Contact	int	10	Not Null
Conductedby	text	50	Not Null
Organizedby	text	50	Not Null
time	varchar	50	Not Null

- camp_enroll

Field Name	Data Type	Size	Constraint
id	int	11	Primary Key, Auto increment
Enroll_camp_name	varchar	20	Not Null
Enroll_name	varchar	50	Not Null
Enroll_number	int	10	Not Null
Enroll_email	varchar	20	Not Null
Enroll_age	int	3	Not Null
Enroll_gender	text	6	Not Null
Enroll_blood	varchar	2	Not Null
Enroll_address	varchar	11	Not Null
Camp_id	int	11	Foreign Key

- **contact_info Table**

Field Name	Data Type	Size	Constraint
contact_id	int	10	Primary Key, Auto increment
contact_address	varchar	50	Not Null
contact_mail	Varchar	50	Not Null
contact_phone	int	10	Not Null

- **pages Table**

Field Name	Data Type	Size	Constraint
page_id	int	10	Primary Key, Auto increment
page_name	varchar	255	Not Null
Page_type	Varchar	50	Not Null
Page_data	longtext	-	Not Null

- **donor_details Table**

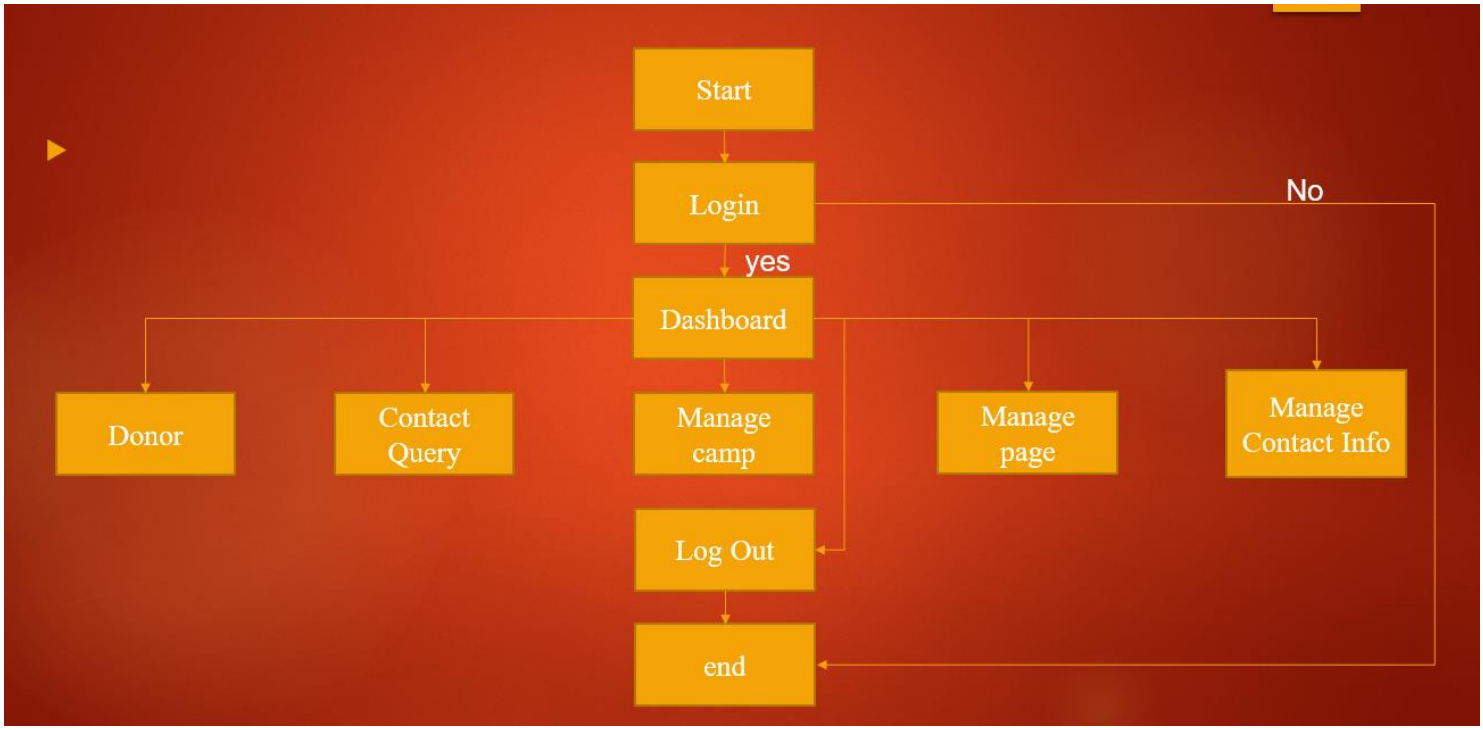
Field Name	Data Type	Size	Constraint
donor_id	int	10	Primary Key, Auto increment
donor_name	text	50	Not Null
Donor_number	Int	10	Not Null
Donor_mail	Varchar	50	Not Null
Donor_age	int	3	Not Null
Donor_gender	varchar	10	Not Null
Donor_blood	varchar	10	Not Null
Donor_address	varchar	100	Not Null

- **contact_query Table**

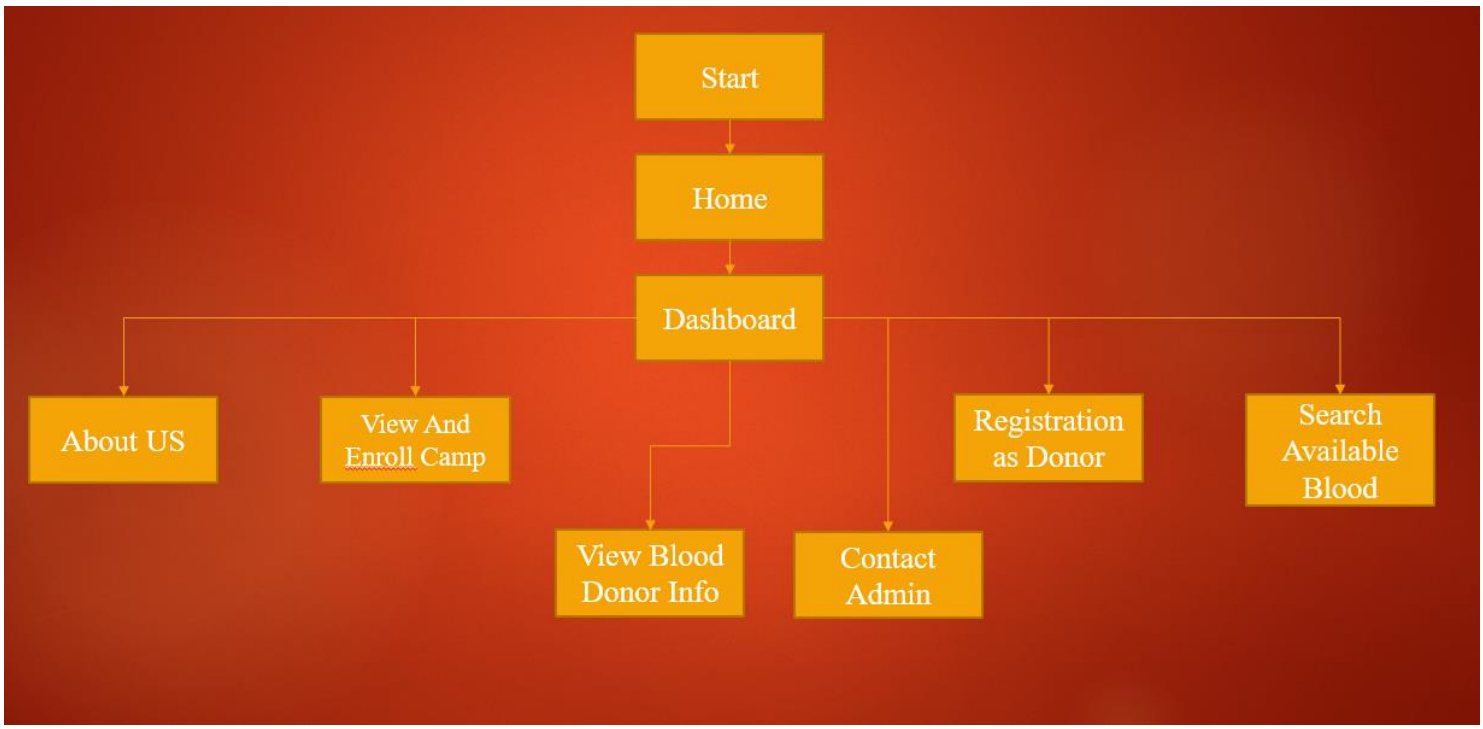
Field Name	Data Type	Size	Constraint
query_id	Int	10	Primary Key, Auto increment
query_name	text	50	Not Null
query_mail	Varchar	100	Not Null
query_number	int	10	Not Null
query_message	longtext	-	Not Null
date	timestamp	50	Not Null

4.3 System Flowchart :

4.1.1 Admin Side :



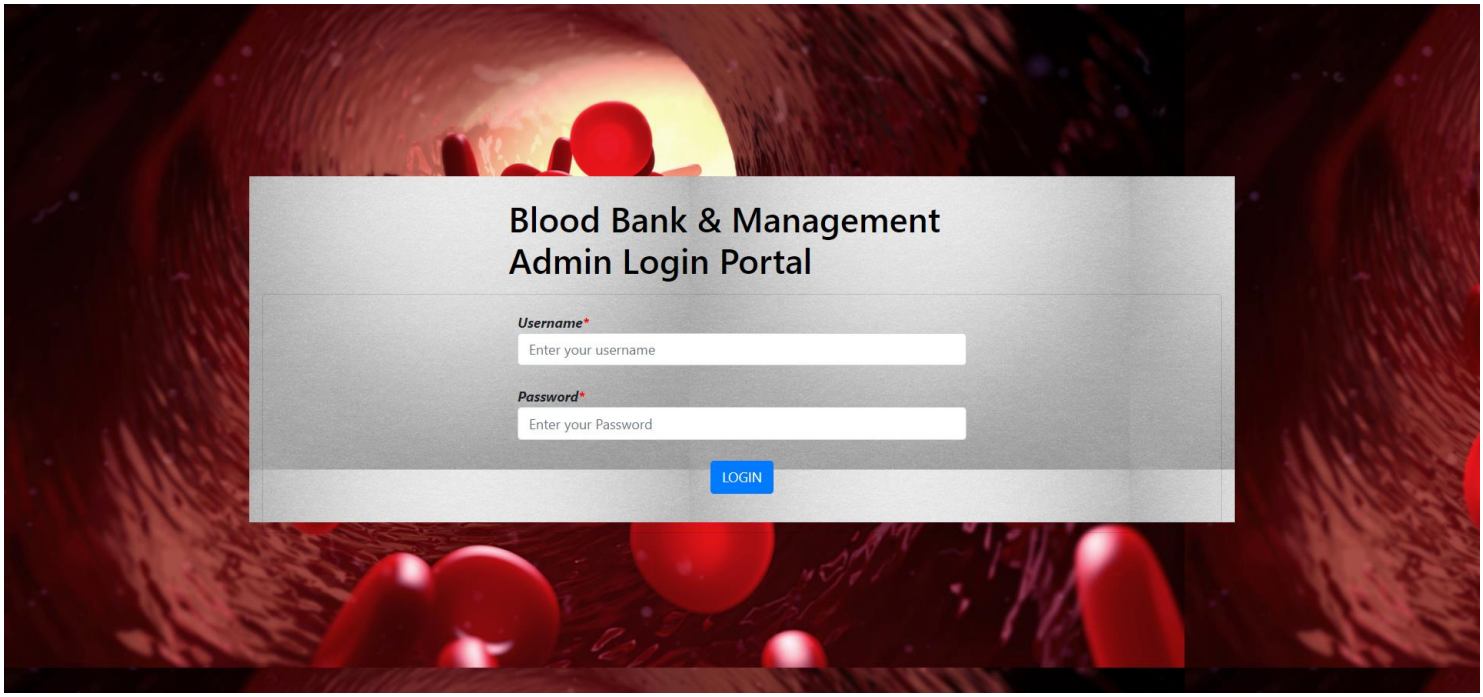
4.1.1 User Side :



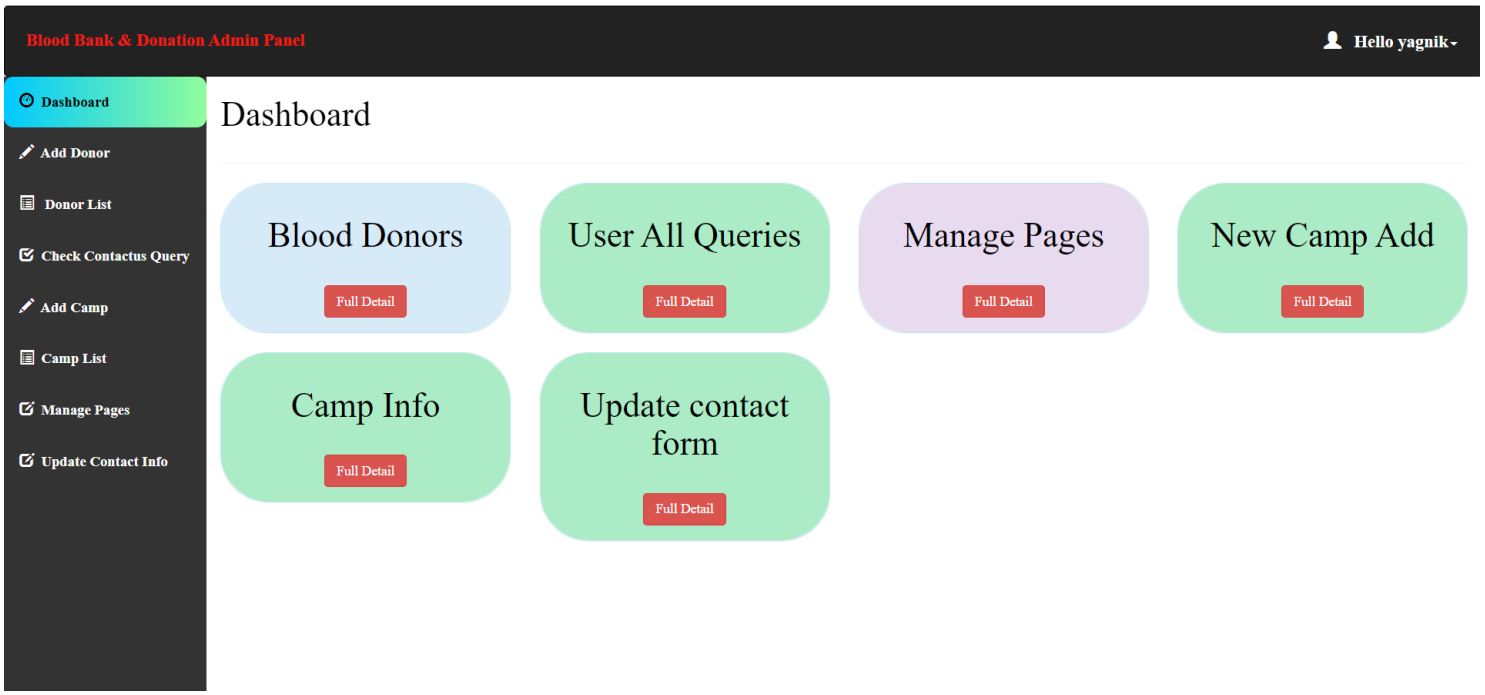
4.4 FORM AND REPORT DESIGN(SCREENSHOT)LAYOUT:

Admin Side :

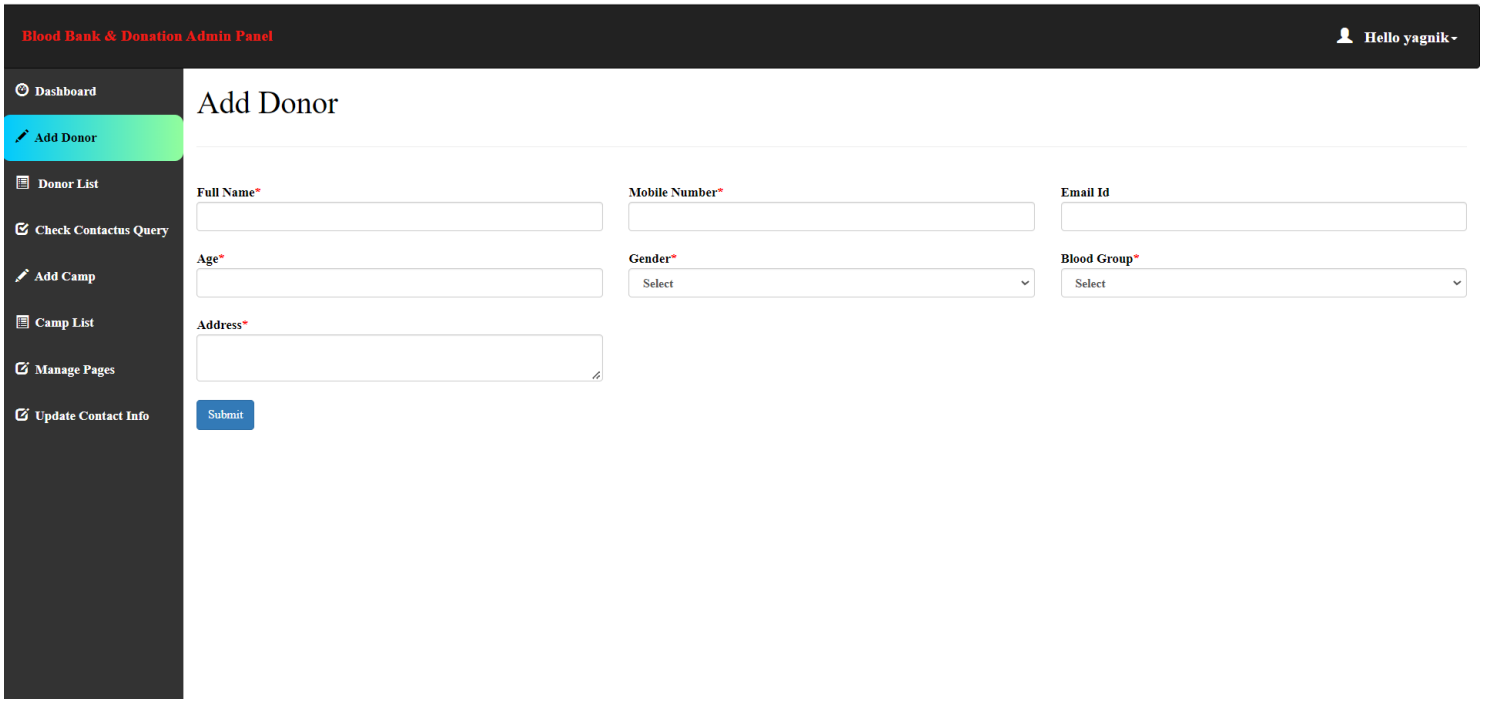
- **Login Page**



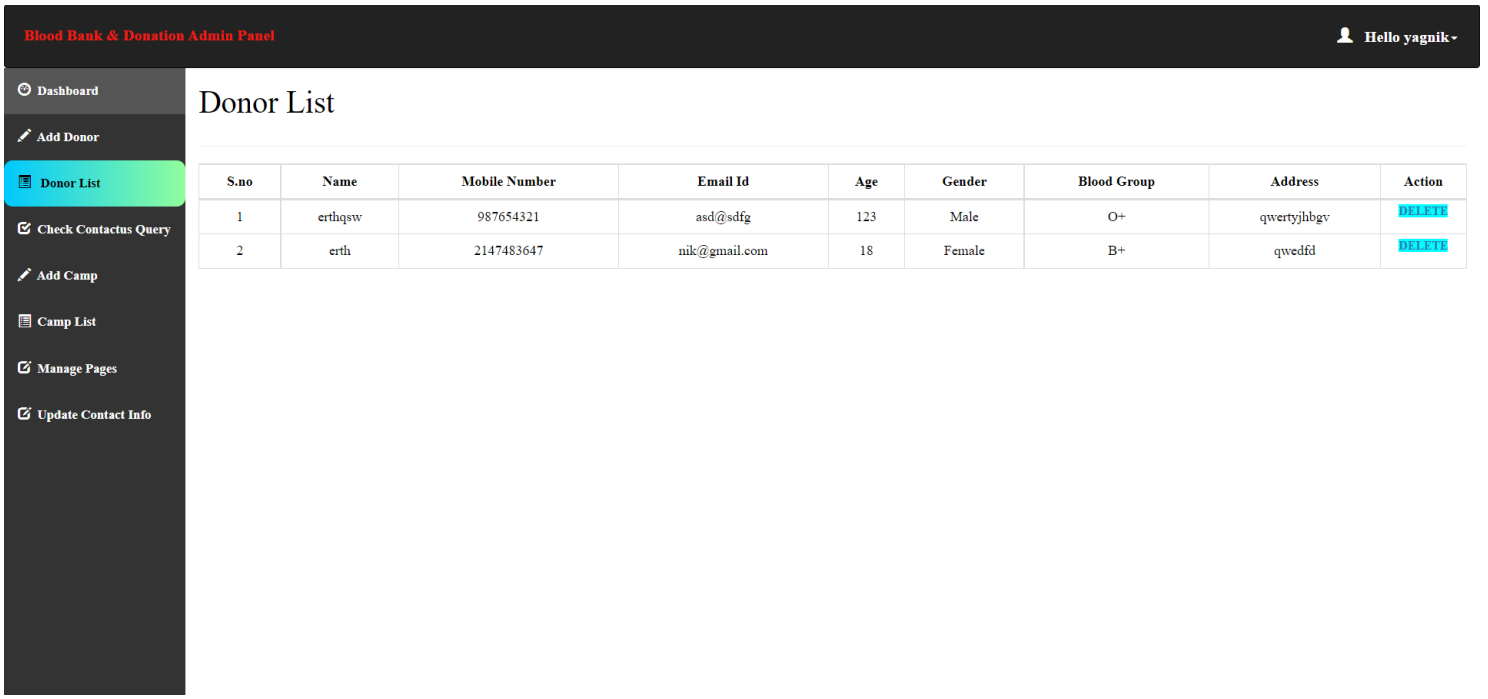
- **Dashboard Page**



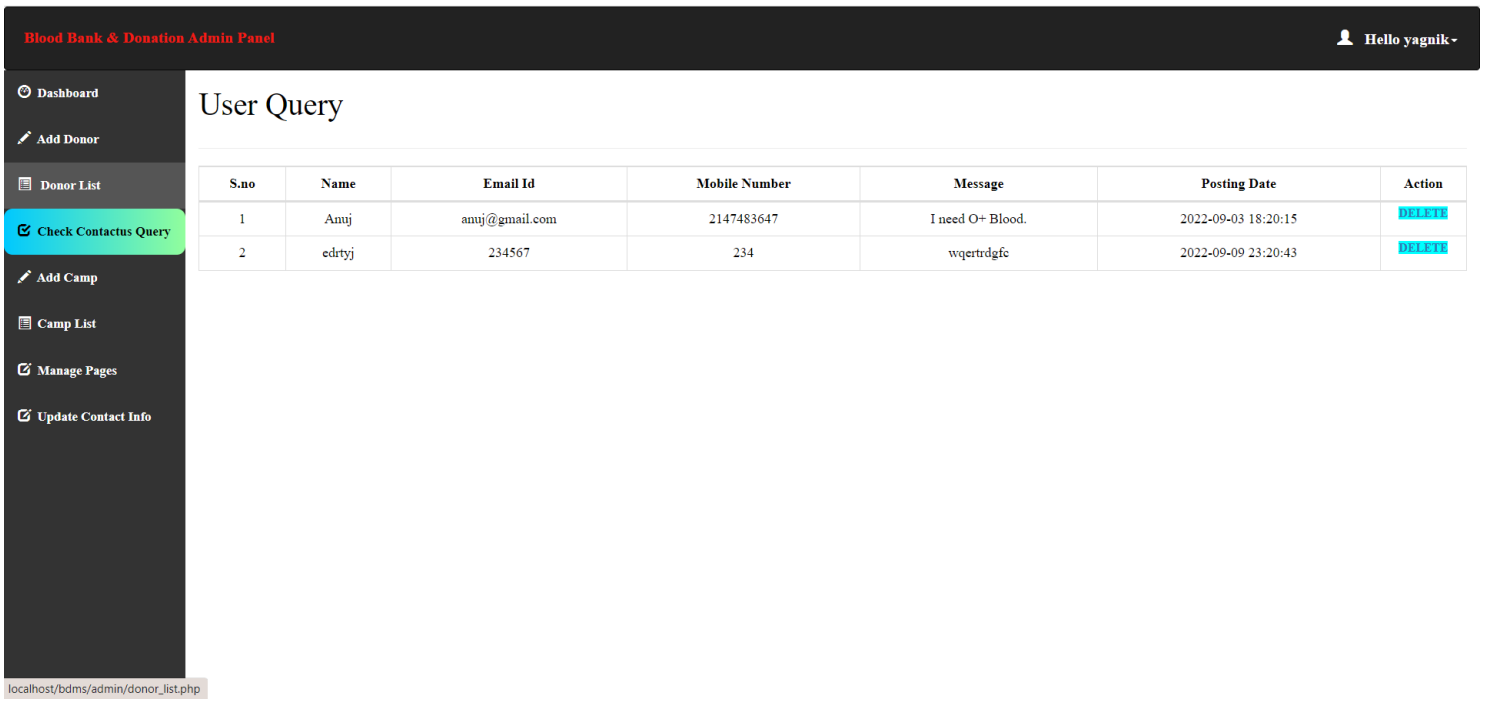
- **Add Donor Page**



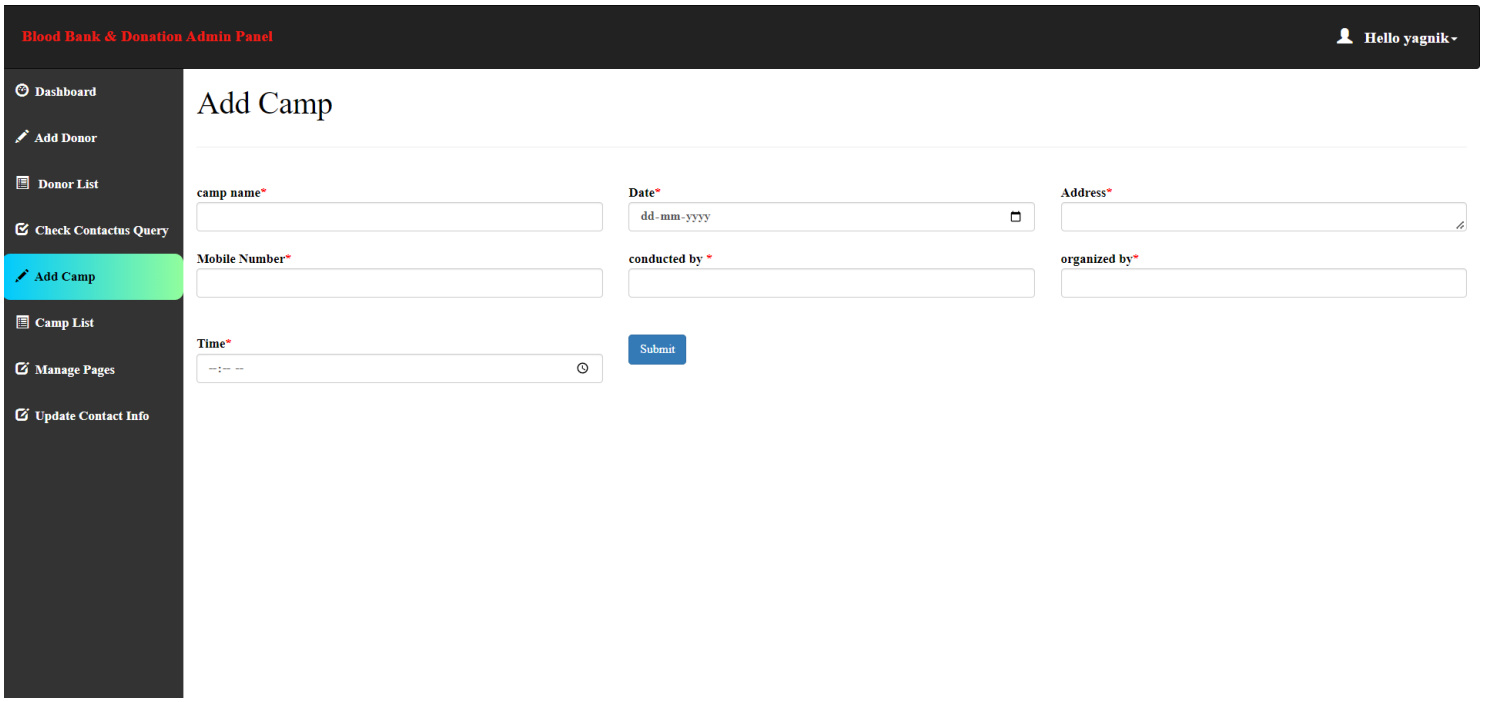
- **Donor List Page**



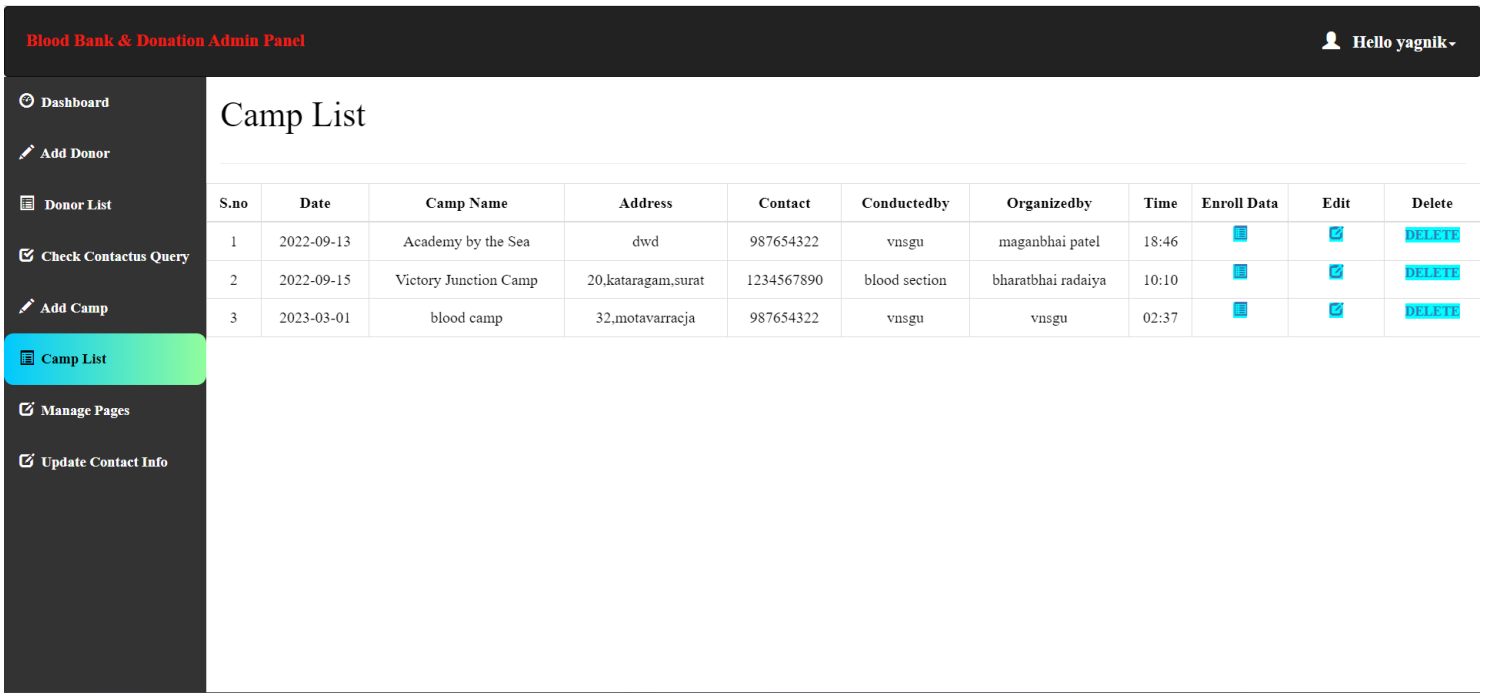
- **User Query Page**



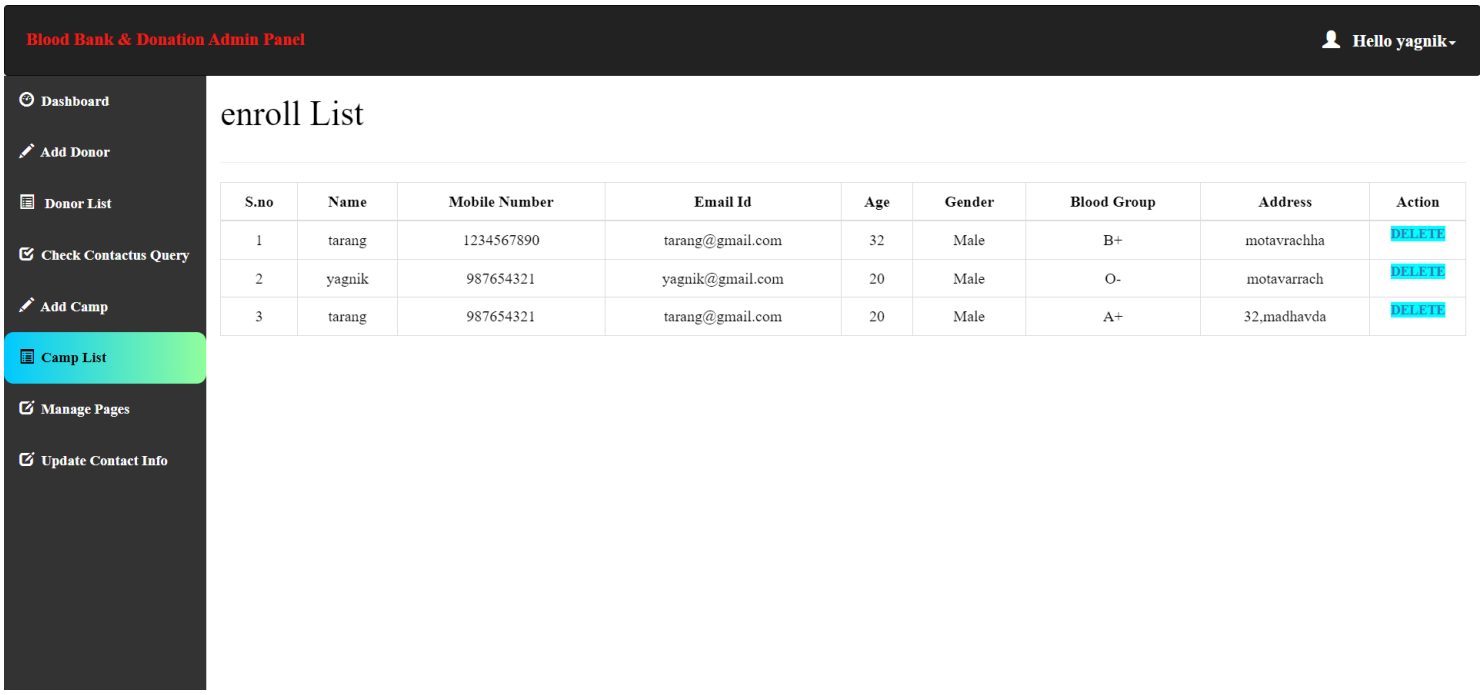
- **Add Camp Page**



- **Camp List Page**

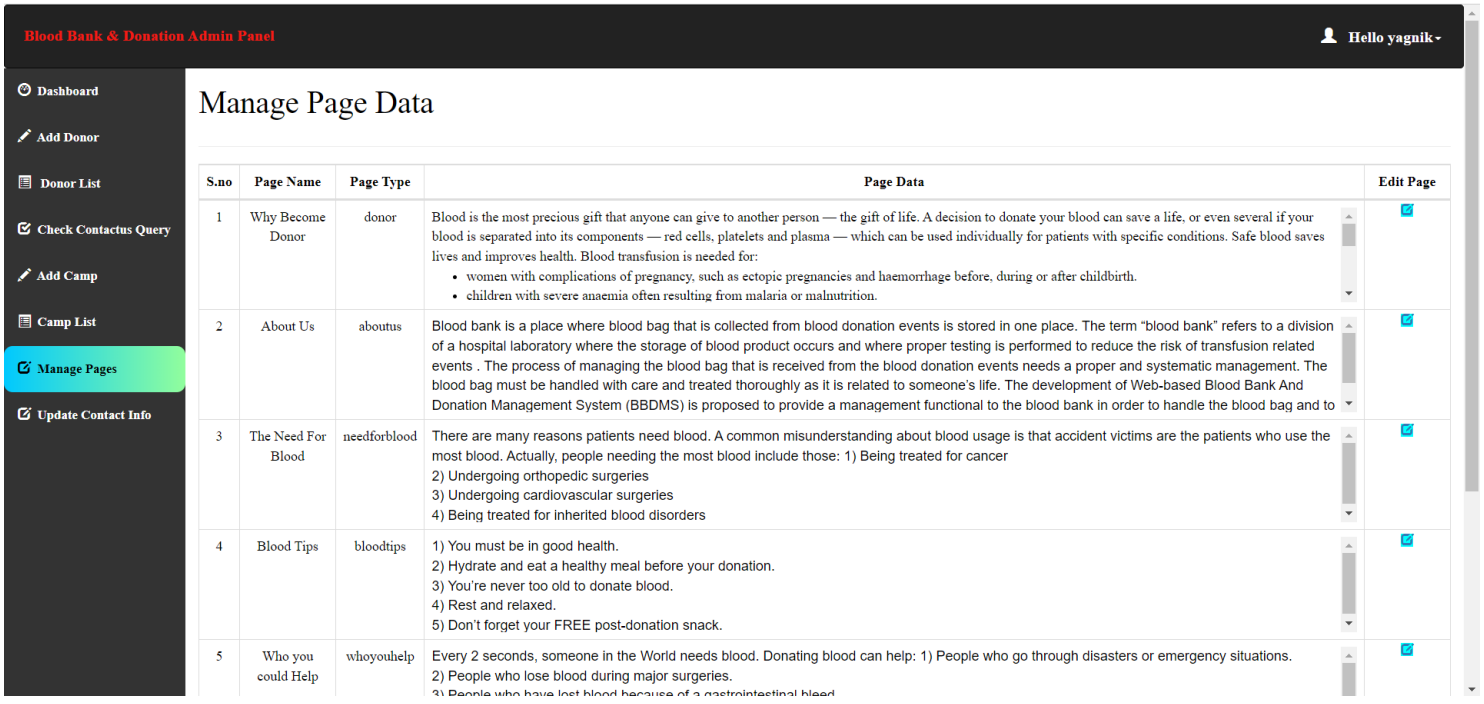


- **Camp Enroll**

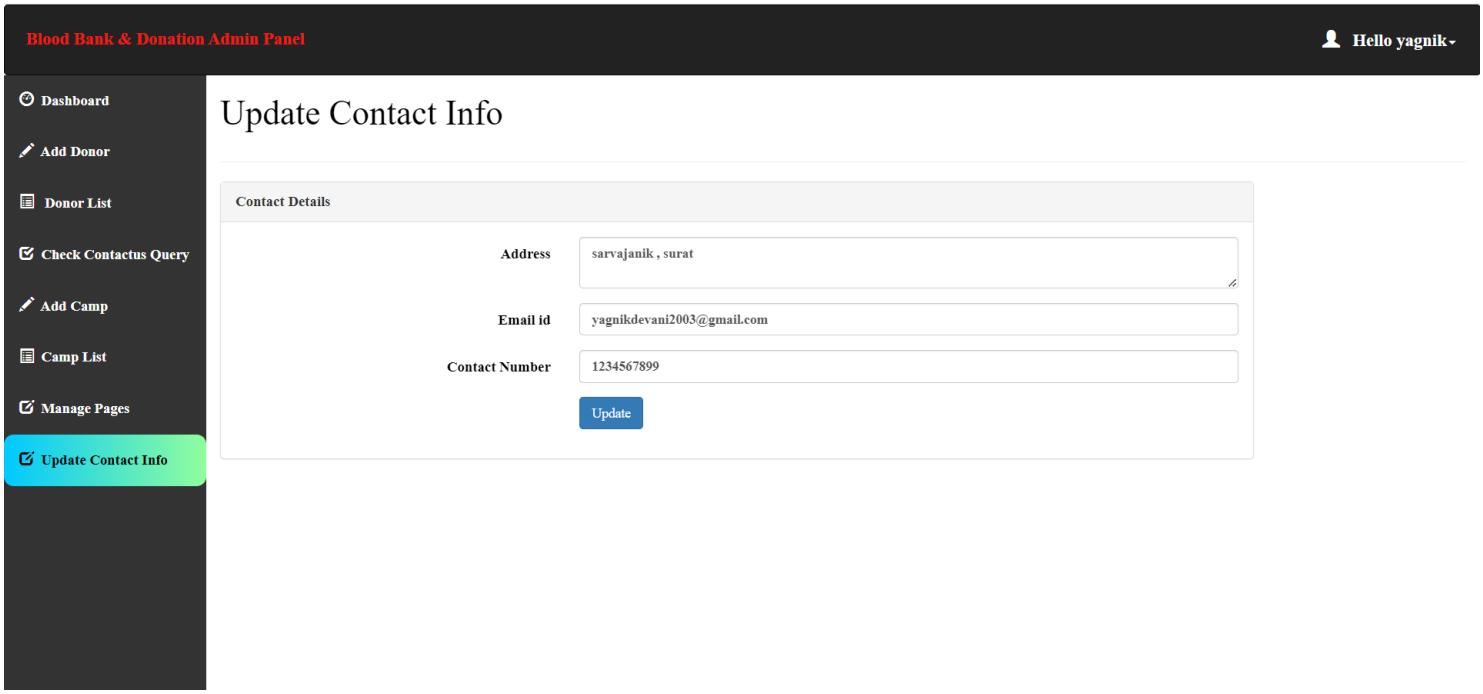


- **Pages Data Manage**

Design

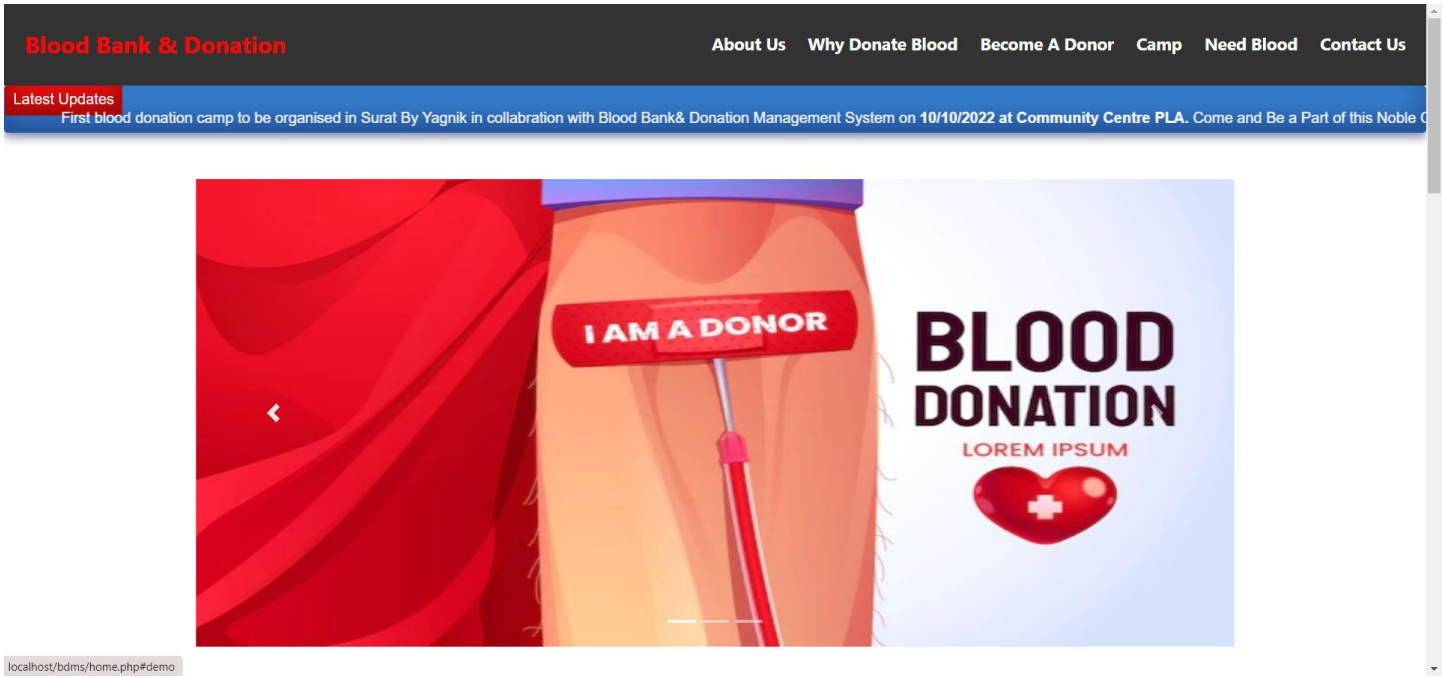


- **Update Admin Contact Info Page**



User Side :

- **Home Page**



Welcome to BloodBank & Donor Management System

The need for blood

There are many reasons patients need blood. A common misunderstanding about blood usage is that accident victims are the patients who use the most

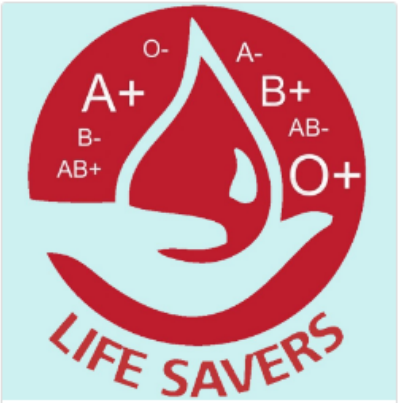
Blood Tips

3) You're never too old to donate blood.
4) Rest and relaxed.
5) Don't forget your FREE post-donation snack.

Who you could Help

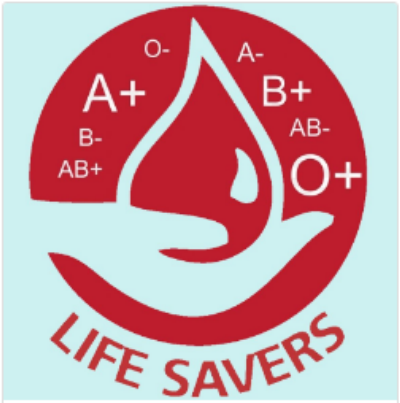
Every 2 seconds, someone in the World needs blood. Donating blood can help: 1) People who go through disasters or emergency situations.

Blood Donor Names



erthqsw

Blood Group : O+
Mobile No. : 987654321
Gender : Male
Age : 123
Address : qwertyjhbgv



erth

Blood Group : B+
Mobile No. : 2147483647
Gender : Female
Age : 18
Address : qwedfd

Camp Notice

S.no	Date	Camp Name	Address	Contact	Conductedby	Organizedby	Time
1	2022-09-13	dgr	dwd	987654322	vnsgu	maganbhai patel	18:46
2	2022-09-15	guyagz	20,kataragam,surat	1234567890	blood section	bharatbhai radaiya	10:10
3	2022-09-15	guyagz	20,kataragam,surat	123456789	blood section	bharatbhai radaiya	10:10
4	2022-09-21	dgr	123456789876	1234567890	vnsgu	sfgh	15:55
5	2022-09-22	dgr	123456543	987654322	fsefg	jenil patel	15:02

BLOOD GROUPS

Blood group of any human being will mainly fall in any one of the following groups.

- A positive or A negative
- B positive or B negative
- O positive or O negative
- AB positive or AB negative.

Your blood group is determined by the genes you inherit from your parents. A healthy diet helps ensure a successful blood donation, and also makes you feel better!



UNIVERSAL DONORS AND RECIPIENTS

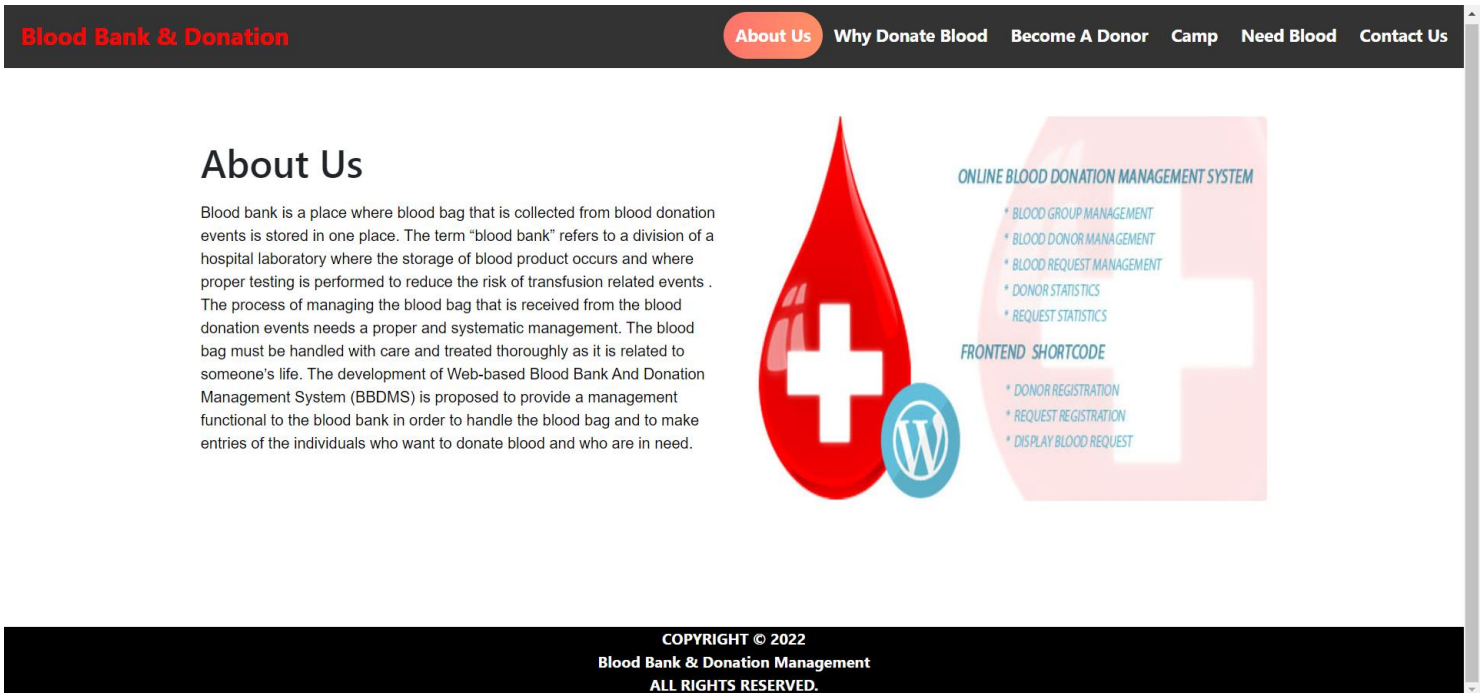
The most common blood type is O, followed by type A. Type O individuals are often called "universal donors" since their blood can be transfused into persons with any blood type. Those with type AB blood are called "universal recipients" because they can receive blood of any type.

For emergency transfusions, blood group type O negative blood is the variety of blood that has the lowest risk of causing serious reactions for most people who receive it. Because of this, it's sometimes called the universal blood donor type.

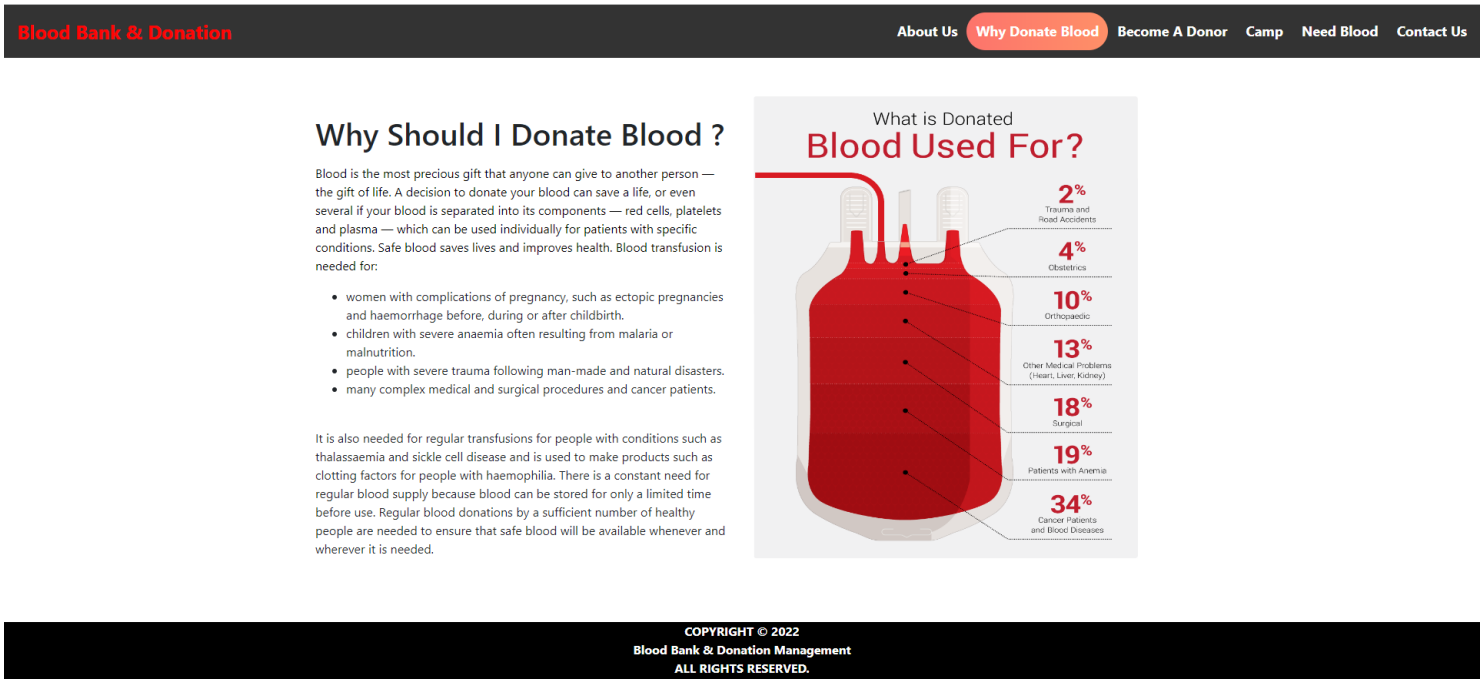
Become a Donor

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Blood Bank & Donation Management
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- **About Us Page**

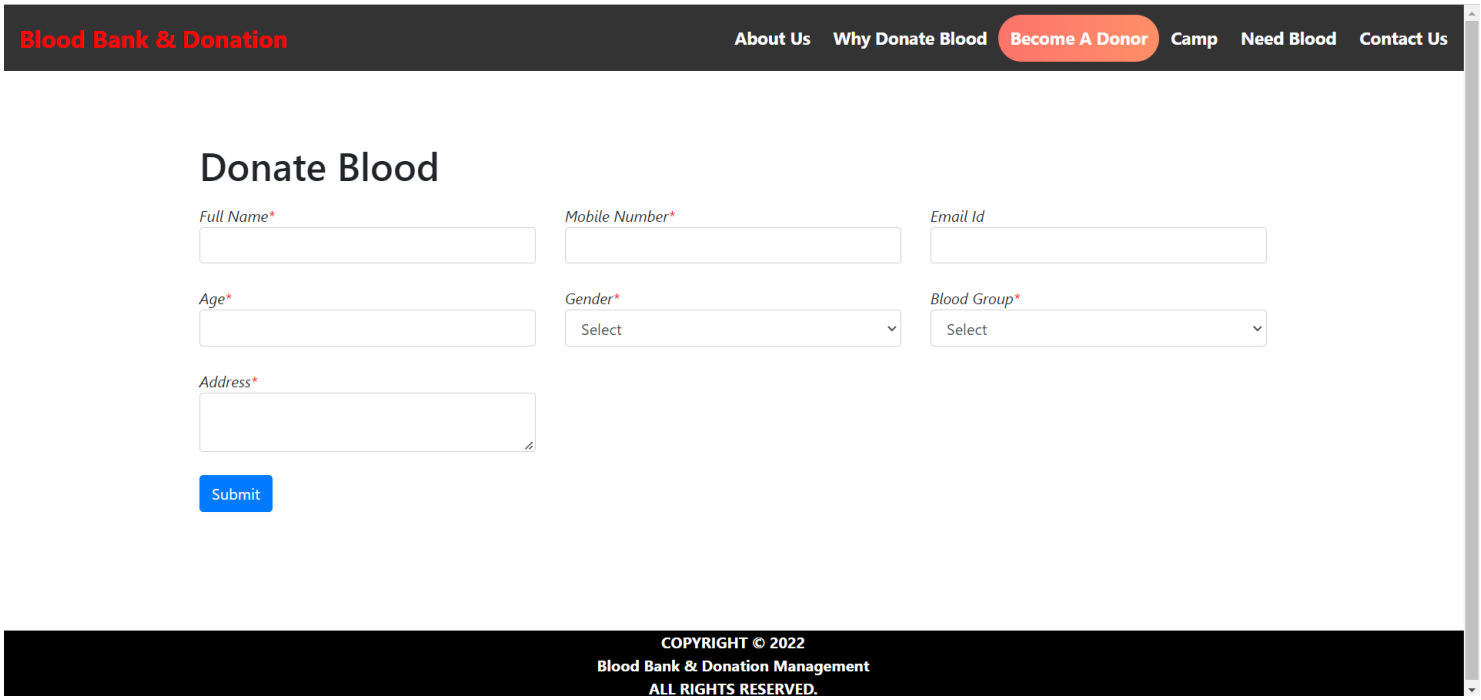


- **Why Donate Blood Page**

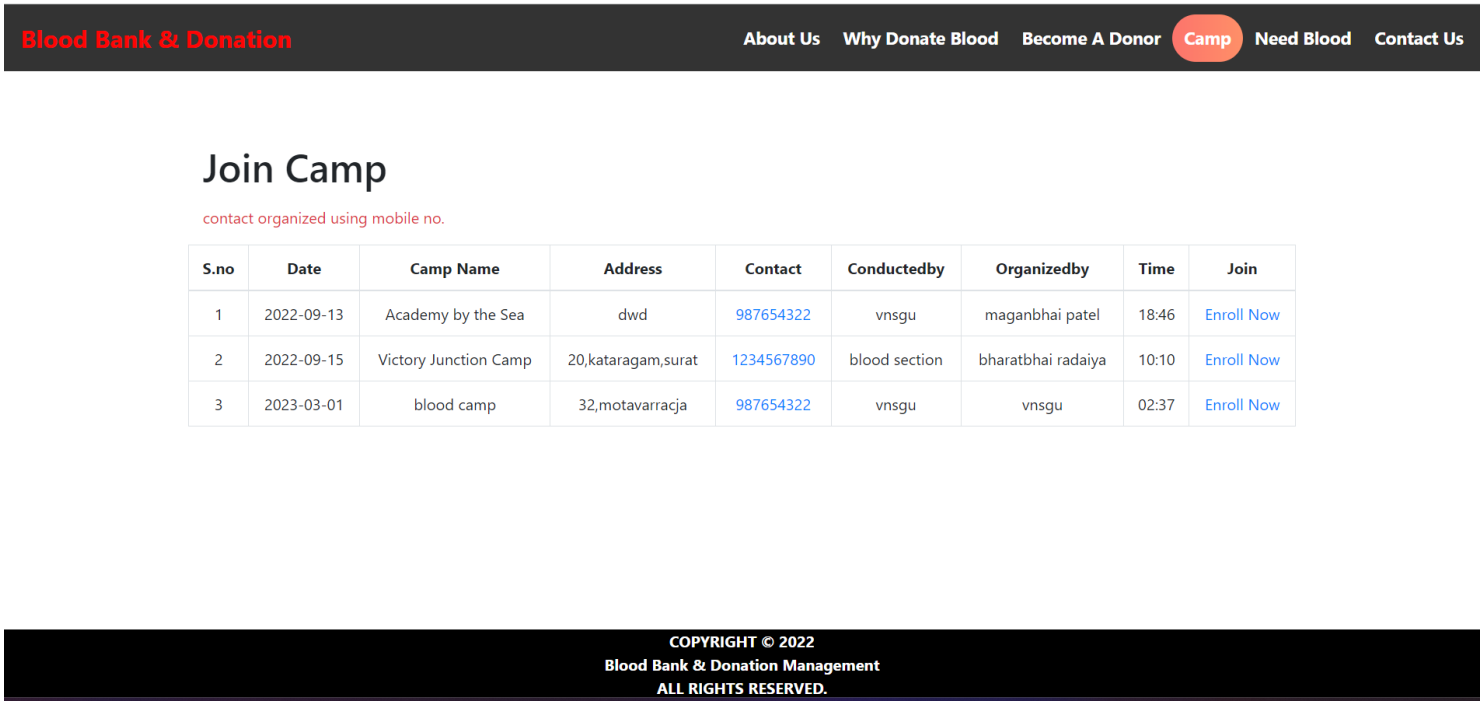


- **Donate Blood Page**

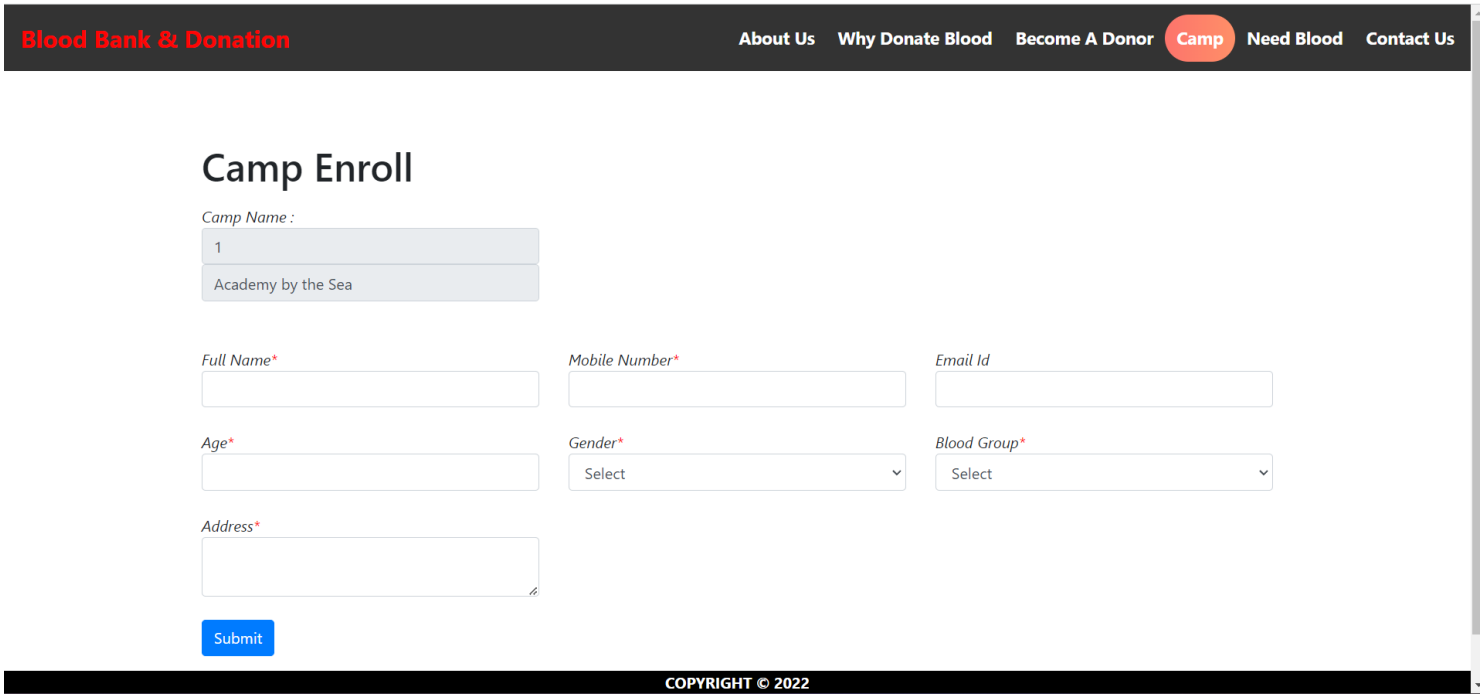
Design



- **Camp Page**

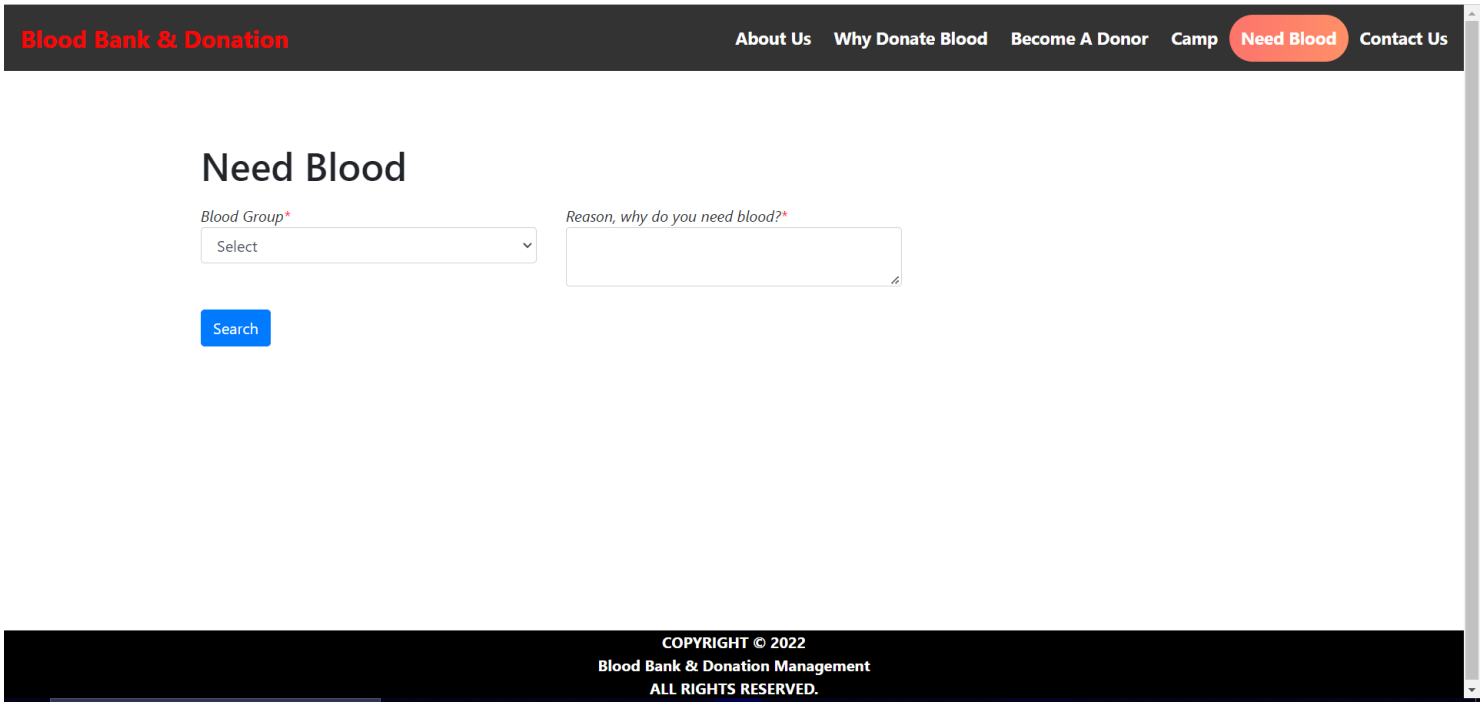


- **Camp Enroll**

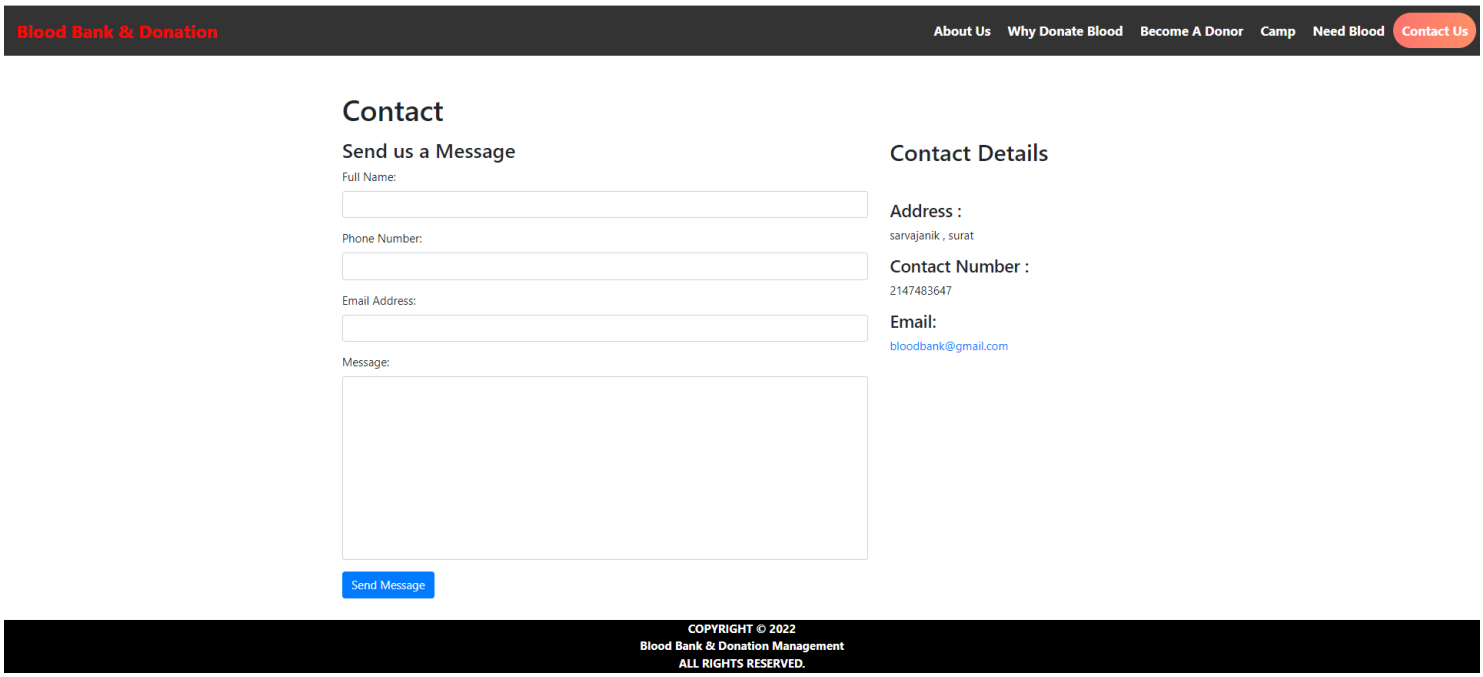


- **Search Blood Page**

Design



- **Admin Contact Query Page**


Blood Bank & Donation
About Us
Why Donate Blood
Become A Donor
Camp
Need Blood
Contact Us
Contact
Send us a Message
Full Name:
Phone Number:
Email Address:
Message:
Send Message
Contact Details
Address :
sarvajanik , surat
Contact Number :
2147483647
Email:
bloodbank@gmail.com
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Blood Bank & Donation Management
ALL RIGHTS RESERVED.

5. Code

User Side :

Donate Blood

```
<html>

<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <meta name="description" content="">
  <meta name="author" content="">
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</head>

<body>
<?php
$active ='donate';
 include('head.php') ?>

<div id="page-container" style="margin-top:50px; position: relative;min-height: 84vh;">
  <div class="container">
  <div id="content-wrap" style="padding-bottom:50px;">
<div class="row">
   <div class="col-lg-6">
      <h1 class="mt-4 mb-3">Donate Blood </h1>
     </div>
</div>
<form name="donor" action="savedata.php" method="post">
```

```
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Full Name<span style="color:red">*</span></div>
<div><input type="text" name="fullname" class="form-control" required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Mobile Number<span style="color:red">*</span></div>
<div><input type="tel" id="phone" name="mobileno" pattern="[0-9]{10}" class="form-control"
required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Email Id</div>
<div><input type="email" name="emailid" class="form-control"></div>
</div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Age<span style="color:red">*</span></div>
<div><input type="number" name="age" class="form-control" min="19" required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Gender<span style="color:red">*</span></div>
<div><select name="gender" class="form-control" required>
<option value="">Select</option>
<option value="Male">Male</option>
<option value="Female">Female</option>
</select>
</div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Blood Group<span style="color:red">*</span></div>
<div><select name="blood" class="form-control" required>
```

Code

```
 <option value=""selected disabled>Select</option>
 <?php
  include 'conn.php';
  $sql= "select * from blood";
  $result=mysqli_query($conn,$sql) or die("query unsuccessful.");
 while($row=mysqli_fetch_assoc($result)){
  ?>
  <option value=" <?php echo $row['blood_id'] ?>"> <?php echo $row['blood_group'] ?> </option>
 <?php } ?>
</select>
</div>
</div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Address<span style="color:red">*</span></div>
<div><textarea class="form-control" name="address" required></textarea></div></div>
</div>
<div class="row">
  <div class="col-lg-4 mb-4">
  <div><input type="submit" name="submit" class="btn btn-primary" value="Submit"
style="cursor:pointer"></div>
  </div>
</div>
</div>
</div>
<?php include('footer.php') ?>
</div>
</body>
</html>
```

Need Blood

```
<html>

<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <meta name="description" content="">
  <meta name="author" content="">
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</head>

<body>
  <?php
  $active ='need';
  include('head.php') ?>

  <div id="page-container" style="margin-top:50px; position: relative;min-height: 84vh;">
    <div class="container">
    <div id="content-wrap" style="padding-bottom:50px;">

  <div class="row">
      <div class="col-lg-6">
          <h1 class="mt-4 mb-3">Need Blood</h1>

        </div>
  </div>
  <form name="needblood" action="" method="post">
  <div class="row">
  <div class="col-lg-4 mb-4">
```

```
<div class="font-italic">Blood Group<span style="color:red">*</span></div>
<div><select name="blood" class="form-control" required>
<option value=""selected disabled>Select</option>
<?php
include 'conn.php';
$sql= "select * from blood";
$result=mysqli_query($conn,$sql) or die("query unsuccessful.");
while($row=mysqli_fetch_assoc($result)){
?>
<option value=" <?php echo $row['blood_id'] ?>"> <?php echo $row['blood_group'] ?> </option>
<?php } ?>
</select>
</div>
</div>

<div class="col-lg-4 mb-4">
<div class="font-italic">Reason, why do you need blood?<span style="color:red">*</span></div>
<div><textarea class="form-control" name="address" required></textarea></div></div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div><input type="submit" name="search" class="btn btn-primary" value="Search"
style="cursor:pointer"></div>
</div>

</div><div class="row">
<?php if(isset($_POST['search'])){

$bg=$_POST['blood'];
$sql= "select * from donor_details join blood where donor_details.donor_blood=blood.blood_id AND
donor_blood='{$bg}' order by rand() limit 5";
$result=mysqli_query($conn,$sql) or die("query unsuccessful.");
```

```
   if(mysqli_num_rows($result)>0)   {
   while($row = mysqli_fetch_assoc($result)) {
     ?>


     <div class="col-lg-4 col-sm-6 portfolio-item" ><br>
     <div class="card" style="width:300px">
        <img class="card-img-top" src="image\blood_drop_logo.jpg" alt="Card image"
style="width:100%;height:300px">
        <div class="card-body">
          <h3 class="card-title"><?php echo $row['donor_name']; ?></h3>
          <p class="card-text">
            <b>Blood Group : </b> <b><?php echo $row['blood_group']; ?></b><br>
            <b>Mobile No. : </b> <?php echo $row['donor_number']; ?><br>
            <b>Gender : </b><?php echo $row['donor_gender']; ?><br>
            <b>Age : </b> <?php echo $row['donor_age']; ?><br>
            <b>Address : </b> <?php echo $row['donor_address']; ?><br>
          </p>


        </div>
       </div>
  </div>


  <?php
    }
  }
    else
    {


       echo '<div class="alert alert-danger">No Donor Found For your search Blood group </div>';


    }
} ?>
```

```
</div>
</div>
</div>
<?php include 'footer.php' ?>
</div>
</body>

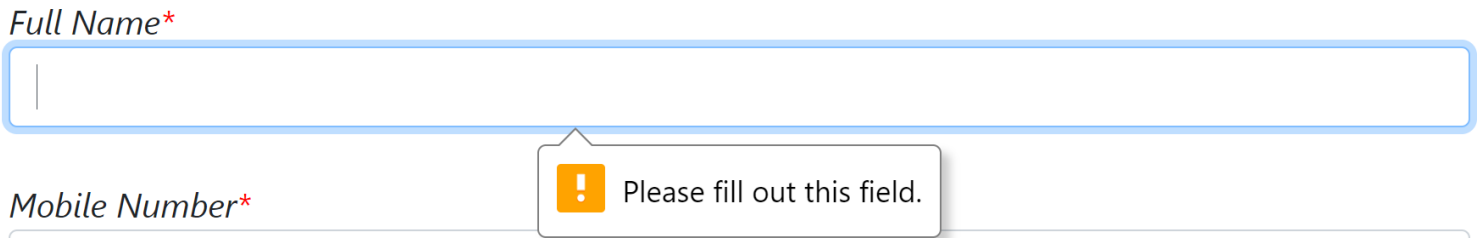
</html>
```

6. Testing

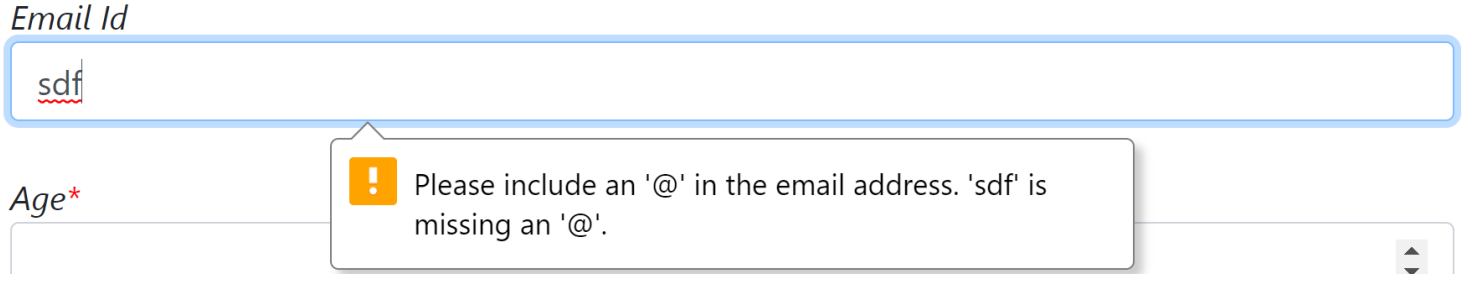
❖ Registration Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty UserName	Blank	Error	Error	Registration Fail
2	Empty Email	Blank/Wrong/Not Valid Formate	Error	Error	Registration Fail
3	Empty Mobile no	Blank/ Require 10 Digit	Error	Error	Registration Fail
4	Empty /18 above age	Blank/Wrong	Error	Error	Registration Fail

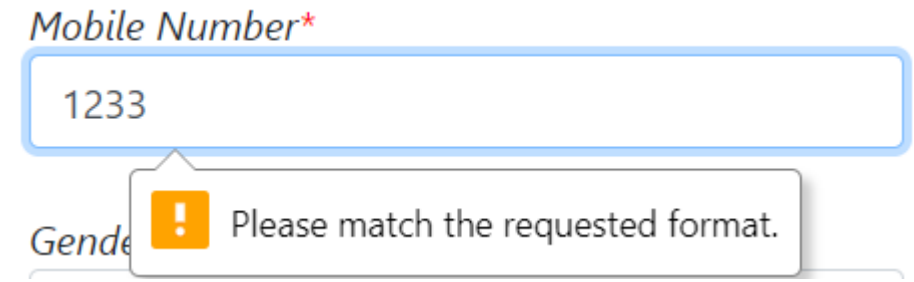
1)



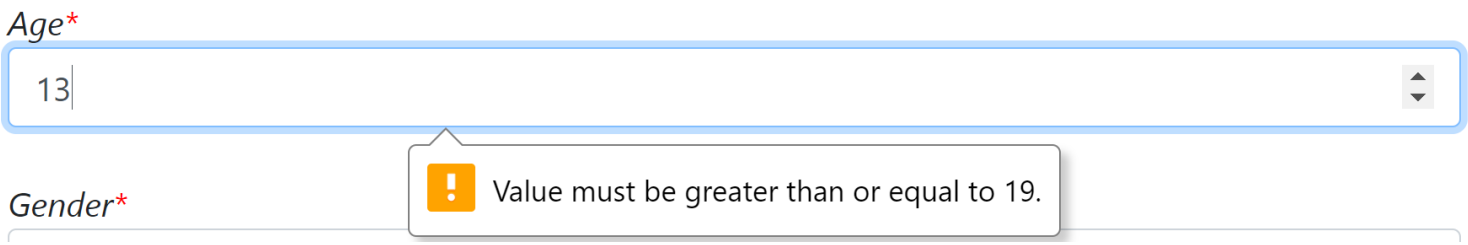
2)



3)


Mobile Number*
1233
Please match the requested format.
Gende

4)


Age*
13
Value must be greater than or equal to 19.
Gender*

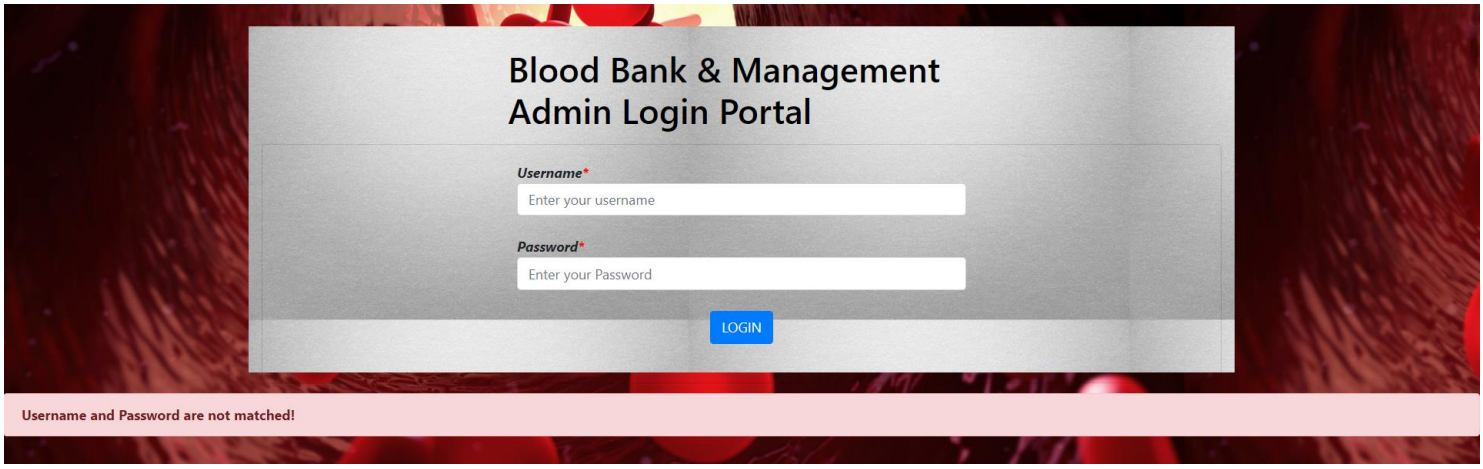
❖ Login Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty UserName	Blank	Error	Error	Login Fail
2	Empty /Wrong password	Blank/Wrong	Error	Error	Login Fail

1)



2)



❖ Camp Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty CampName	Blank	Error	Error	Camp Registration Fail
2	Empty Date	Blank	Error	Error	Camp Registration Fail
3	Empty Mobile no	Blank Require 10 Digit	Error	Error	Camp Registration Fail
4	Empty /Address	Blank	Error	Error	Camp Registration Fail
5	Empty /Time	Blank	Error	Error	Camp Registration Fail
6	Empty /conducted /organized	Blank	Error	Error	Camp Registration Fail

camp name*

Date*

dd-mm-yyyy

Address*

Mobile Number*

conducted by *

organized by*

Time*

--:-- --

Submit

7.0 NEW TOOLS/ TECHNOLOGIES LEARNED/ USED:

- HTML
- CSS
- JavaScript
- Bootstrap
- PHP
- My SQL Database (PHP My Admin)

•

8.0 SYSTEM LIMITATION/CONSTRAINTS:

- User can not get email
- Admin not add directly admin add by other admin
- Camp contact using website

9.Future Enhancement & Opportunities

- More efficient design and user experience.
- Apply More efficient constraints .
- Add module to admin can change password using Mobile No./Email OTP.
- Add feature of login/signup using google / facebook.

10. BIBLIOGRAPHY & REFERENCES:

Books

- PHP: Learn PHP in 24 Hours or Less - A Beginner's Guide To Learning PHP Programming Now

Website

- www.stackoverflow.com
- www.javatpoint.com
- www.tutorialspoint.com